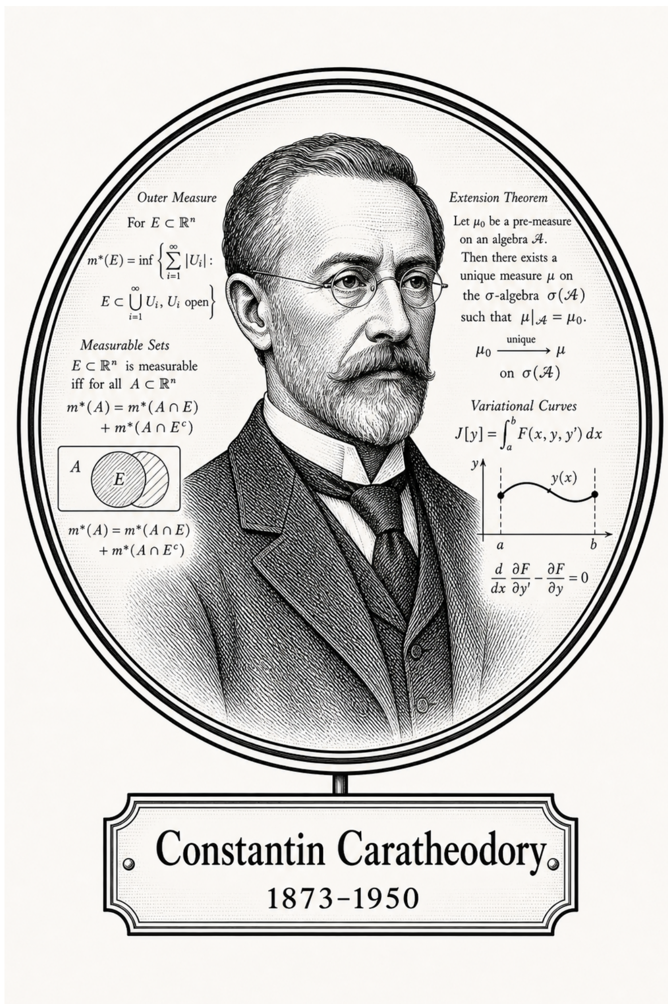
From Cantor to Ito

Mathematical Spaces



Constantin Caratheodory

1873-1950

Self-Study Notes

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For every reader learning to make mathematics their own.

Contents

Part I

Mathematical Spaces

Chapter 1

Mathematical Spaces

1.1 Spaces as Structured Arenas

Definition — Space

Definition 1.1.1 (Space)**.** A **space** is a nonempty set.

Remark (Standard quantified statement)*.*

$$X \text{ is a space} \iff X \neq \varnothing.$$

Remark (Interpretation)*.* At the base level, the word "space" records that there is an arena of points under discussion. No distance, topology, operation, or measure has yet been chosen. Later phrases such as metric space, topological space, and measure space add structure to this underlying nonempty set.

Remark (Historical note)*.* This convention appears as Definition 2.1 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Exposition)*.* A mathematical space is an arena in which a particular kind of mathematics can be performed. At the base there is a set of objects. Added structure then specifies the ground rules: which objects can be compared, which subsets are distinguished, which operations are allowed, which limits make sense, or which quantities can be measured. The same underlying set can support many different spaces because different structures make different questions meaningful.

Remark (Structural Pattern)*.* The general pattern is

$$\text{space} = \text{underlying set} + \text{chosen structure}.$$

A metric space chooses a distance function. A topological space chooses a collection of open sets. A measurable space chooses a σ-algebra of measurable sets, and a measure space adds a measure to that measurable structure. Algebraic spaces choose operations and laws. Each choice controls which statements are legitimate inside that setting.

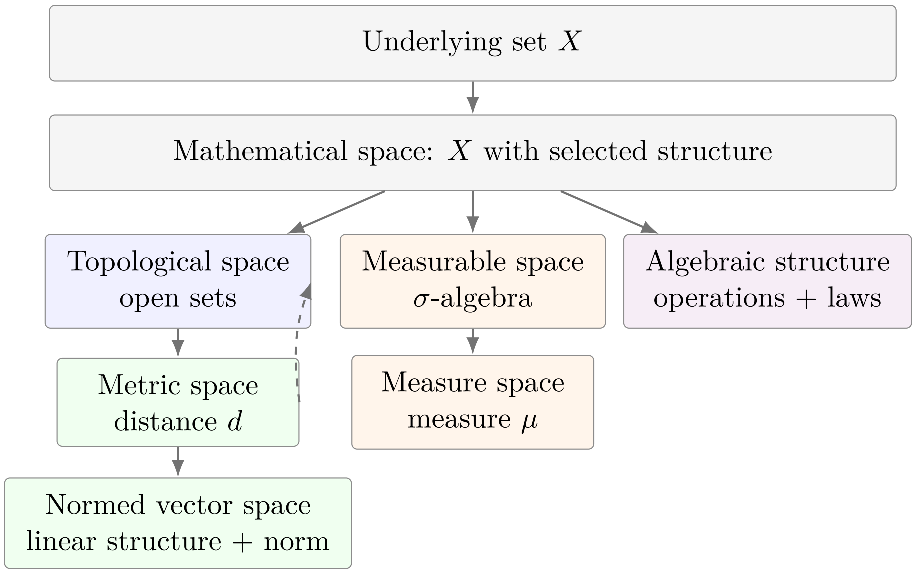
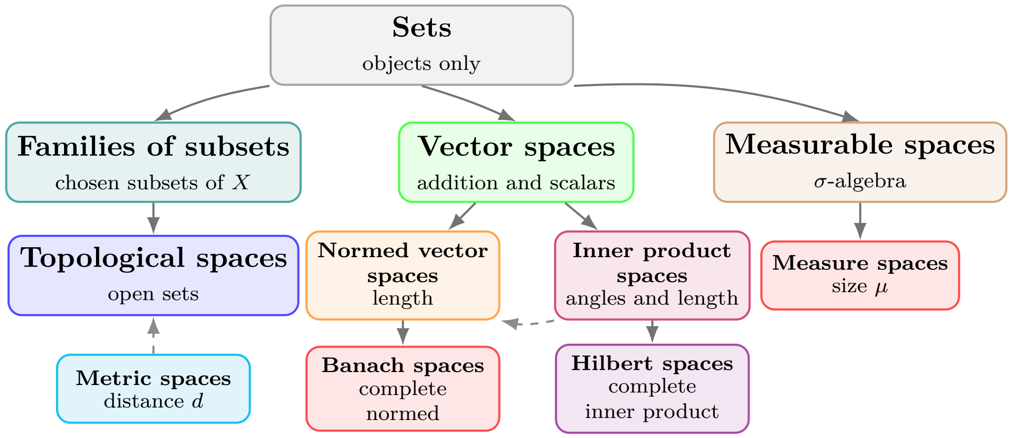


Figure 1.1: Mathematical spaces are built by adding structure to an underlying set. Some structures refine others: a metric induces a topology, while a measure requires measurable sets.

Remark (Interpretation). The nesting in Figure 1.1 should be read as an increase in available structure, not as a claim that every subject lies on a single chain. Metric spaces sit naturally over topology because every metric determines open balls and hence open sets. Measure spaces follow a different branch: they organize size rather than nearness. The common feature is the same: mathematical work begins only after the arena and its rules have been specified.

Remark (Exposition). The hierarchy in Figure 1.2 is a guide to structure, not a universal taxonomy. Some spaces are formed by adding data to a set. Others are formed by adding compatibility requirements to an already structured space. The following table records the construction pattern that will be used throughout this volume.

Remark (Structure Table).



The branches record added structure. A topology selects open sets, a metric supplies distance and induces open sets, linear spaces add operations, and measure spaces add measurable sets together with size.

Figure 1.2: One common route from sets to structured spaces. A topology begins with selected open subsets of a set. A metric gives a distance and thereby generates open balls, so metric spaces naturally carry topologies. Linear, normed, inner product, Banach, and Hilbert spaces add further compatible structure.

Space	**Comes from**	**Additional structure or rules**
Set	Primitive arena	A collection of objects; no nearness, size, or operation is specified.
Family of subsets	Set X	A selected collection $\mathcal{A} \subseteq \mathcal{P}(X)$.
Topological space	Set X	A topology $\mathcal{T}$: open sets containing $\varnothing$ and X, closed under arbitrary unions and finite intersections.
Metric space	Set X	A metric $d : X \times X \to [0, \infty)$ satisfying identity of indiscernibles, symmetry, and the triangle inequality; the metric induces open balls and hence a topology.
Vector space	Set V	Addition and scalar multiplication satisfying the vector space axioms over a chosen field.
Normed vector space	Vector space	A norm $\lVert\cdot\rVert$ measuring vector length, compatible with scalar multiplication and addition.
Inner product space	Vector space	An inner product $\langle \cdot, \cdot \rangle$ measuring angles and lengths; it induces a norm.
Banach space	Normed vector space	Completeness: every Cauchy sequence converges in the norm.
Hilbert space	Inner product space	Completeness with respect to the norm induced by the inner product.
Measurable space	Set X	A σ-algebra $\mathcal{M}$ of subsets regarded as measurable.
Measure space	Measurable space	A measure $\mu : \mathcal{M} \to [0, \infty]$ assigning size and satisfying countable additivity.

Remark (Structural Cards). The following cards use the structural presentation layer: each card records the inherited arena, the new data or laws, and the role of the resulting space. These are blueprint artifacts, not formal definitions; formal definitions remain authoritative when they are introduced.

Structural Card: Set

TAXONOMY

Primitive arena.

NEW AT

A nonempty domain X of objects.

AXIOMS / THEORY

No nearness, size, operation, or distinguished family of subsets is specified.

Structural Card: Family of Subsets

INHERITS

Set X.

NEW AT

A chosen collection $\mathcal{A} \subseteq \mathcal{P}(X)$.

TAXONOMY

Subset-selection structure. No closure law is imposed at this level.

Structural Card: Topological Space

INHERITS

Set X with a distinguished family of subsets.

NEW AT

A topology $\mathcal{T} \subseteq \mathcal{P}(X)$, read as the open subsets of X.

AXIOMS / THEORY

$\varnothing, X \in \mathcal{T}$; arbitrary unions of members of $\mathcal{T}$ lie in $\mathcal{T}$; finite intersections of members of $\mathcal{T}$ lie in $\mathcal{T}$.

TAXONOMY

Open-set structure for continuity, convergence, interior, closure, and neighborhood arguments.

Structural Card: Metric Space

INHERITS

Set X.

NEW AT

A distance function

$$d : X \times X \to [0, \infty).$$

AXIOMS / THEORY

Identity of indiscernibles, symmetry, and the triangle inequality.

USES-A

Open balls $B(x, r) = \{y \in X : d(x, y) < r\}$.

TAXONOMY

Distance structure. The metric induces a topology by declaring sets open when they contain a metric ball around each of their points.

Structural Card: Vector Space

INHERITS

Set V and a scalar field $\mathbb{F}$.

NEW AT

Vector addition $+ : V \times V \to V$, scalar multiplication $\cdot : \mathbb{F} \times V \to V$, and a zero vector $0 \in V$.

AXIOMS / THEORY

$(V, +, 0)$ is an abelian group; scalar multiplication is unital and associative; scalar multiplication distributes over vector addition and field addition.

TAXONOMY

Linear structure. The space supports linear combinations before any length, angle, or limit structure is chosen.

Structural Card: Normed Vector Space

INHERITS

Vector space V over $\mathbb{F}$.

NEW AT

A norm $\|\cdot\| : V \to [0, \infty)$.

AXIOMS / THEORY

$\|v\| = 0$ if and only if $v = 0$; $\|\alpha v\| = |\alpha| \, \|v\|$; $\|u + v\| \le \|u\| + \|v\|$.

USES-A

The induced metric $d(u, v) = \|u - v\|$.

TAXONOMY

Linear metric structure. Algebraic operations and metric notions become compatible.

Structural Card: Inner Product Space

INHERITS

Real or complex vector space V.

NEW AT

An inner product $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{F}$.

AXIOMS / THEORY

Positive definiteness, conjugate symmetry, and linearity in the chosen argument.

USES-A

The induced norm $\|v\| = \sqrt{\langle v, v \rangle}$.

TAXONOMY

Linear geometry. Length, angle, and orthogonality are available, and the metric structure is derived from the inner product.

Structural Card: Banach Space

INHERITS

Normed vector space $(V, \|\cdot\|)$.

NEW AT

Completeness with respect to the norm-induced metric.

AXIOMS / THEORY

Every Cauchy sequence in V converges to a point of V in the norm.

TAXONOMY

Complete normed linear space. Approximation by Cauchy sequences remains inside the space.

Structural Card: Hilbert Space

INHERITS

Inner product space H.

NEW AT

Completeness with respect to the norm induced by the inner product.

AXIOMS / THEORY

Every Cauchy sequence in the induced norm converges to a point of H.

TAXONOMY

Complete inner product space. Orthogonality and projection arguments take place in a complete metric environment.

Structural Card: Measurable Space

INHERITS

Set X with a distinguished family of subsets.

NEW AT

A σ-algebra $\mathcal{M} \subseteq \mathcal{P}(X)$, read as the measurable subsets of X.

AXIOMS / THEORY

$X \in \mathcal{M}$; complements of measurable sets are measurable; countable unions of measurable sets are measurable.

TAXONOMY

Countably closed subset-selection structure. It specifies which subsets are legal inputs for size assignments.

Structural Card: Measure Space

INHERITS

Measurable space $(X, \mathcal{M})$.

NEW AT

A measure $\mu : \mathcal{M} \to [0, \infty]$.

AXIOMS / THEORY

$\mu(\varnothing) = 0$, and μ is countably additive on pairwise disjoint measurable sets:

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n).$$

TAXONOMY

Quantitative measurable structure. The σ-algebra determines the admissible subsets, and the measure assigns their size.

Chapter 2

Metric Spaces

SERIES	VOLUME IV	BOOK I	CHAPTER 2
From Cantor to Ito	Mathematical Spaces	Mathematical Spaces	Metric Spaces

2.1 Metric Spaces

2.2 Definitions

Definition 2.2.1 (Metric Space)**.** Let X be a nonempty set. A function

$$d : X \times X \to \mathbb{R}_{\geq 0}$$

is called a *metric* on X if for all $x, y, z \in X$,

1. $d(x, y) \geq 0$, and $d(x, y) = 0$ if and only if $x = y$ *(positive definiteness)*;
2. $d(x, y) = d(y, x)$ *(symmetry)*;
3. $d(x, y) \leq d(x, z) + d(z, y)$ *(triangle inequality)*.

A *metric space* is a pair (X, d), where X is a nonempty set and d is a metric on X.

Remark (Standard quantified statement).

$$\begin{aligned} \mathrm{Metric}(X, d) \iff\ & X \neq \varnothing \ \wedge\ d : X \times X \to \mathbb{R}_{\geq 0} \\ & \wedge \forall x, y, z \in X \colon d(x, y) \geq 0 \wedge \big(d(x, y) = 0 \iff x = y\big) \\ & \wedge d(x, y) = d(y, x) \wedge d(x, y) \leq d(x, z) + d(z, y). \end{aligned}$$

Remark (Interpretation). A metric turns a set into a distance arena. Positive definiteness says that distance is never negative and that zero distance identifies exactly one point. Symmetry says that distance does not depend on direction. The triangle inequality says that moving through an intermediate point cannot make the direct distance larger than the two-step route.

Remark (Historical note). This convention appears as Definition 2.2 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Definition 2.2.2 (Pseudo-Metric)**.** Let X be a nonempty set. A function

$$d : X \times X \to \mathbb{R}_{\geq 0}$$

is called a *pseudo-metric* on X if for all $x, y, z \in X$,

1. $d(x, y) \geq 0$, and $d(x, x) = 0$ *(positive semi-definiteness)*;
2. $d(x, y) = d(y, x)$ *(symmetry)*;
3. $d(x, y) \leq d(x, z) + d(z, y)$ *(triangle inequality)*.

Remark (Standard quantified statement).

$$\begin{aligned} \mathrm{PseudoMetric}(X, d) \iff\ & X \neq \varnothing \ \wedge \ d : X \times X \to \mathbb{R}_{\geq 0} \\ & \wedge \forall x, y, z \in X \colon d(x, y) \geq 0 \wedge d(x, x) = 0 \\ & \wedge d(x, y) = d(y, x) \wedge d(x, y) \leq d(x, z) + d(z, y). \end{aligned}$$

Remark (Interpretation). A pseudo-metric keeps the distance-like laws of nonnegativity, symmetry, and the triangle inequality, but it weakens positive definiteness. Distinct points may have distance 0. Thus a pseudo-metric measures separation only up to possible zero-distance identifications.

Remark (Historical note). This convention appears as Definition 2.6 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Lemma

Lemma 2.2.3 (Metrics are Pseudo-Metrics)**.** *Every metric on a nonempty set X is a pseudo-metric on X. The converse is false: there exist pseudo-metrics that are not metrics.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &\forall X\, \forall d\colon\ \mathrm{Metric}(X, d) \Longrightarrow \mathrm{PseudoMetric}(X, d), \\ &\exists X\, \exists d\colon\ \mathrm{PseudoMetric}(X, d) \wedge \neg\, \mathrm{Metric}(X, d). \end{aligned}$$

Remark (Interpretation). The implication is immediate because the metric axioms contain all pseudo-metric axioms. The only difference is the zero-distance law: a metric forces $d(x, y) = 0$ exactly when $x = y$, while a pseudo-metric only requires $d(x, x) = 0$.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Pseudo-Metric

2.3 Metrics

Remark (Exposition). A metric is the distance function itself. A metric space is the set together with that function. This section records standard metric functions and then states, one by one, that each function satisfies the metric axioms.

2.3.1 Norms Induce Metrics

Definition — Norm

Definition 2.3.1 (Norm)**.** Let V be a real vector space. A *norm* on V is a function

$$\|\cdot\| : V \to \mathbb{R}_{\geq 0}$$

such that for all $x, y \in V$ and all $\lambda \in \mathbb{R}$,

1. $\|x\| = 0$ if and only if $x = 0$ *(positive definiteness)*;
2. $\|\lambda x\| = |\lambda| \, \|x\|$ *(absolute homogeneity)*;
3. $\|x + y\| \leq \|x\| + \|y\|$ *(triangle inequality)*.

Remark (Standard quantified statement).

$$\begin{aligned} \mathrm{Norm}(V, \|\cdot\|) \iff & \|\cdot\| : V \to \mathbb{R}_{\geq 0} \\ & \wedge \forall x \in V\colon \|x\| = 0 \iff x = 0 \\ & \wedge \forall \lambda \in \mathbb{R} \; \forall x \in V\colon \|\lambda x\| = |\lambda| \, \|x\| \\ & \wedge \forall x, y \in V\colon \|x + y\| \leq \|x\| + \|y\|. \end{aligned}$$

Remark (Interpretation). A norm measures the size of one vector. The metric induced by a norm measures the size of the displacement vector $x - y$. Thus a norm is a length structure, and the induced metric is the corresponding distance structure.

Theorem

Theorem 2.3.2 (A Norm Induces a Metric)**.** *Let V be a real vector space and let $\|\cdot\|$ be a norm on V. Define*

$$d_{\|\cdot\|} : V \times V \to \mathbb{R}_{\geq 0}, \qquad d_{\|\cdot\|}(x, y) := \|x - y\|.$$

Then $d_{\|\cdot\|}$ is a metric on V.
Go to proof.

Remark (Standard quantified statement).

$$\mathrm{Norm}(V, \|\cdot\|) \implies \mathrm{Metric}(V, d_{\|\cdot\|}), \qquad d_{\|\cdot\|}(x, y) = \|x - y\|.$$

Remark (Interpretation). The metric axioms are exactly the norm axioms applied to displacement vectors. Zero distance says the displacement $x - y$ has zero norm, hence is the zero vector. Symmetry follows from absolute homogeneity with scalar -1. The triangle inequality follows by writing $x - y = (x - z) + (z - y)$.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Norm

Definition — Discrete Metric

Definition 2.3.3 (Discrete Metric)**.** Let X be a nonempty set. The *discrete metric* on X is the function

$$d_{\text{disc}} : X \times X \to \mathbb{R}_{\geq 0}$$

defined by

$$d_{\text{disc}}(x, y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases}$$

Remark (Standard quantified statement).

$$\begin{aligned} &X \neq \varnothing, && d_{\text{disc}} : X \times X \to \mathbb{R}_{\geq 0}, \\ &\forall x, y \in X\colon && d_{\text{disc}}(x, y) = 0 \iff x = y, \\ &\forall x, y \in X\colon && x \neq y \iff d_{\text{disc}}(x, y) = 1. \end{aligned}$$

Remark (Interpretation). The discrete metric treats equality as the only closeness relation. Distinct points are all placed at the same positive distance from each other.

Proposition

Proposition 2.3.4 (The Discrete Distance is a Metric)**.** *For every nonempty set X, the discrete distance d_{disc} is a metric on X.*

Go to proof.

Remark (Standard quantified statement).

$$X \neq \varnothing \implies \mathrm{Metric}(X, d_{\text{disc}}).$$

Remark (Interpretation). The only point at distance 0 from x is x itself. Symmetry is built into the two-case definition, and the triangle inequality is automatic because the only possible distances are 0 and 1.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Discrete Metric

Definition — Euclidean Distance on the Real Line

Definition 2.3.5 (Euclidean Distance on $\mathbb{R}$)**.** The *Euclidean distance* on $\mathbb{R}$ is the function

$$d_{\mathbb{R}} : \mathbb{R} \times \mathbb{R} \to \mathbb{R}_{\geq 0}, \qquad d_{\mathbb{R}}(x, y) = |x - y|.$$

Remark (Standard quantified statement).

$$d_{\mathbb{R}} : \mathbb{R} \times \mathbb{R} \to \mathbb{R}_{\geq 0}, \qquad \forall x, y \in \mathbb{R}: \quad d_{\mathbb{R}}(x, y) = |x - y|.$$

Remark (Interpretation). The real-line distance records absolute displacement between two real numbers. It forgets direction and keeps only separation.

Proposition

Proposition 2.3.6 (The Real-Line Euclidean Distance is a Metric)**.** *The function $d_{\mathbb{R}}(x, y) = |x - y|$ is a metric on $\mathbb{R}$.*
Go to proof.

Remark (Standard quantified statement).

$$\mathrm{Metric}(\mathbb{R}, d_{\mathbb{R}}).$$

Remark (Interpretation). This proposition says that the usual absolute-value distance satisfies the abstract metric axioms.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Euclidean Distance on $\mathbb{R}$

Definition — Euclidean Space

Definition 2.3.7 (Euclidean Space)**.** For $n \in \mathbb{N}$, the *n-dimensional Euclidean space* is $\mathbb{R}^n$ equipped with its standard coordinate structure, dot product, Euclidean norm, and Euclidean distance.

Remark (Standard quantified statement). For each $n \in \mathbb{N}$, $\mathrm{EuclideanSpace}(n)$ denotes $\mathbb{R}^n$ together with its standard coordinate structure, dot product, Euclidean norm, and Euclidean distance.

Remark (Interpretation). Euclidean space is the coordinate model of ordinary distance geometry.

Remark (Historical note). This convention is the setting for the Euclidean examples in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Definition — Euclidean Distance on $\mathbb{R}^n$

Definition 2.3.8 (Euclidean Distance on $\mathbb{R}^n$)**.** For $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_n)$ in $\mathbb{R}^n$, the *Euclidean distance* is the function

$$d_2 : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}_{\geq 0}, \qquad d_2(x, y) = \sqrt{(x_1 - y_1)^2 + \cdots + (x_n - y_n)^2}.$$

Remark (Standard quantified statement).

$$d_2 : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}_{\geq 0}, \qquad d_2(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}.$$

Remark (Interpretation). The Euclidean distance measures the length of the coordinate difference vector $x - y$.

Theorem

Theorem 2.3.9 (Cauchy–Schwarz Inequality). *For every $x, y \in \mathbb{R}^n$,*

$$|x \cdot y| \leq |x|\,|y|.$$

Equality holds if and only if x and y are linearly dependent.

Go to proof.

Remark (Standard quantified statement).

$$\forall x, y \in \mathbb{R}^n: \quad |x \cdot y| \leq |x|\,|y|.$$

Remark (Interpretation). Cauchy–Schwarz controls the dot product by the lengths of the two vectors. It is the algebraic estimate behind Euclidean triangle inequalities.

Remark (Dependencies).

- Definition: Euclidean Space
- Definition: Euclidean Distance on $\mathbb{R}^n$

Corollary

Corollary 2.3.10 (Minkowski's Inequality). *For every $x, y \in \mathbb{R}^n$,*

$$|x + y| \leq |x| + |y|.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x, y \in \mathbb{R}^n: \quad |x + y| \leq |x| + |y|.$$

Remark (Interpretation). Minkowski's inequality is the triangle inequality for Euclidean length.

Remark (Dependencies).

- Theorem: Cauchy–Schwarz Inequality
- Definition: Euclidean Distance on $\mathbb{R}^n$

Proposition

Proposition 2.3.11 (The Euclidean Distance on $\mathbb{R}^n$ is a Metric)**.** *The function d_2 is a metric on $\mathbb{R}^n$.*
Go to proof.

Remark (Standard quantified statement).

$$\mathrm{Metric}(\mathbb{R}^n, d_2).$$

Remark (Interpretation). The Euclidean distance makes $\mathbb{R}^n$ into the standard finite dimensional metric space.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Euclidean Distance on $\mathbb{R}^n$
- Corollary: Minkowski's Inequality

Corollary

Corollary 2.3.12 (Euclidean Metric)**.** *For every $x, y \in \mathbb{R}^n$, define*

$$d_2(x, y) := \sqrt{|x_1 - y_1|^2 + \cdots + |x_n - y_n|^2}.$$

Then $(\mathbb{R}^n, d_2)$ is a metric space.
Go to proof.

Remark (Standard quantified statement).

$$\mathrm{Metric}(\mathbb{R}^n, d_2).$$

Remark (Interpretation). This records the standard Euclidean metric-space structure on $\mathbb{R}^n$.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Euclidean Distance on $\mathbb{R}^n$
- Corollary: Minkowski's Inequality

Definition — Taxicab Metric

Definition 2.3.13 (Taxicab Metric)**.** For $x, y \in \mathbb{R}^n$, the *taxicab metric* is the function

$$d_1(x, y) := \sum_{i=1}^{n} |x_i - y_i|.$$

Remark (Standard quantified statement).

$$d_1 : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}_{\geq 0}, \qquad d_1(x, y) = \sum_{i=1}^{n} |x_i - y_i|.$$

Remark (Interpretation). Taxicab distance measures coordinate displacement additively, as if travel were constrained to coordinate-parallel streets.

Proposition

Proposition 2.3.14 (Taxicab Metric). *The function d_1 is a metric on $\mathbb{R}^n$.*
Go to proof.

Remark (Standard quantified statement).

$$\mathrm{Metric}(\mathbb{R}^n, d_1).$$

Remark (Interpretation). The taxicab metric satisfies the metric axioms coordinate by coordinate, with the real absolute-value triangle inequality summed over all coordinates.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Taxicab Metric

Definition — p-Metric on $\mathbb{R}^n$

Definition 2.3.15 (p-Metric on $\mathbb{R}^n$). Let $1 \leq p < \infty$. For $x, y \in \mathbb{R}^n$, define

$$d_p(x, y) := \left(\sum_{i=1}^{n} |x_i - y_i|^p \right)^{1/p}.$$

Remark (Standard quantified statement).

$$1 \leq p < \infty \implies d_p : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}_{\geq 0}.$$

Remark (Interpretation). The p-metrics interpolate between taxicab distance at $p = 1$ and Euclidean distance at $p = 2$, and they form the standard finite-dimensional ℓ^p family.

Theorem

Theorem 2.3.16 (d_p is a Metric). *For every $1 \leq p < \infty$, the function d_p is a metric on $\mathbb{R}^n$.*
Go to proof.

Remark (Standard quantified statement).

$$1 \leq p < \infty \implies \mathrm{Metric}(\mathbb{R}^n, d_p).$$

Remark (Interpretation). The proof is the p-norm version of Minkowski's inequality. The theorem packages a whole family of metric structures on the same underlying set.

Remark (Dependencies).

- Definition: Metric Space
- Definition: p-Metric on $\mathbb{R}^n$

Definition — Complex Metric

Definition 2.3.17 (Complex Metric)**.** Identify $\mathbb{C}$ with $\mathbb{R}^2$. The *complex metric* is

$$d_{\mathbb{C}}(z, w) := |z - w|.$$

Remark (Standard quantified statement).

$$d_{\mathbb{C}} : \mathbb{C} \times \mathbb{C} \to \mathbb{R}_{\geq 0}, \qquad d_{\mathbb{C}}(z, w) = |z - w|.$$

Remark (Interpretation). As a distance function, the complex metric is the Euclidean metric on $\mathbb{R}^2$ written in complex notation.

Proposition

Proposition 2.3.18 (The Complex Plane is a Metric Space)**.** *The pair* $(\mathbb{C}, d_{\mathbb{C}})$ *is a metric space.*

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{Metric}(\mathbb{C}, d_{\mathbb{C}}).$$

Remark (Interpretation). The metric on $\mathbb{C}$ is inherited from the Euclidean plane.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Complex Metric
- Corollary: Euclidean Metric

Definition — Supremum Metric on Bounded Real Sequences

Definition 2.3.19 (Supremum Metric on Bounded Real Sequences)**.** Let $\ell^\infty(\mathbb{R})$ denote the set of all bounded real sequences:

$$\ell^\infty(\mathbb{R}) := \{x = (x_n)_{n \in \mathbb{N}} : \exists M \geq 0 \; \forall n \in \mathbb{N} \; |x_n| \leq M\}.$$

For $x, y \in \ell^\infty(\mathbb{R})$, define

$$d_\infty(x, y) := \sup_{n \in \mathbb{N}} |x_n - y_n|.$$

Remark (Standard quantified statement).

$$d_\infty : \ell^\infty(\mathbb{R}) \times \ell^\infty(\mathbb{R}) \to \mathbb{R}_{\geq 0}, \qquad d_\infty(x,y) = \sup_{n\in\mathbb{N}} |x_n - y_n|.$$

Remark (Interpretation). The supremum metric measures the largest coordinatewise discrepancy between two bounded sequences.

Proposition

Proposition 2.3.20 (The Supremum Distance on Bounded Sequences is a Metric)**.** *The function d_∞ is a metric on $\ell^\infty(\mathbb{R})$.*
Go to proof.

Remark (Standard quantified statement).

$$\mathrm{Metric}(\ell^\infty(\mathbb{R}), d_\infty).$$

Remark (Interpretation). Two bounded sequences are close in this metric when every coordinate is close by the same tolerance.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Supremum Metric on Bounded Real Sequences

2.3.2 Point Functions and Pointlike Functions

Remark (Exposition). A metric is a two-variable distance function

$$d : X \times X \to \mathbb{R}_{\geq 0}.$$

The codomain $\mathbb{R}_{\geq 0}$ records only a nonnegative distance value; by itself, it does not remember which point of X served as the reference point. Point functions adapt the metric by freezing one input. For a fixed $z \in X$, the function $\delta_z : X \to \mathbb{R}_{\geq 0}$ sends x to the distance $d(z,x)$. Thus a point of the metric space can be represented by its distance profile across the whole space.

Definition — Point Function

Definition 2.3.21 (Point Function)**.** Let (X,d) be a metric space and let $z \in X$. The *point function at z* is the function

$$\delta_z : X \to \mathbb{R}_{\geq 0}, \qquad \delta_z(x) := d(z,x).$$

The set of all point functions on X is denoted by

$$\delta(X) := \{\, \delta_z : z \in X \,\}.$$

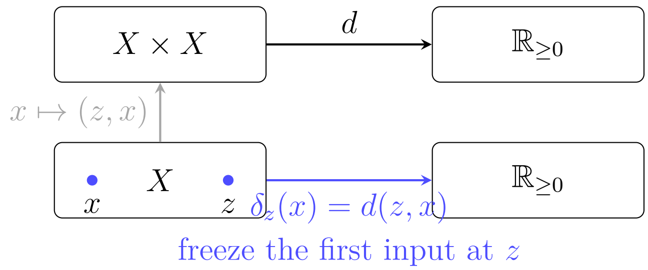


Figure 2.1: A metric $d : X \times X \to \mathbb{R}_{\geq 0}$ becomes the point function $\delta_z : X \to \mathbb{R}_{\geq 0}$ when the first input is fixed at z.

Remark (Standard quantified statement).

$$z \in X \implies \delta_z : X \to \mathbb{R}_{\geq 0} \quad \text{and} \quad \forall x \in X\colon \delta_z(x) = d(z, x).$$

Remark (Interpretation). The point function δ_z is the distance-from-z profile. It turns the fixed point z into a one-variable nonnegative real-valued function on the whole space.

Remark (Examples). In $\mathbb{R}$ with the usual metric $d(x, y) = |x - y|$, the point function at $z \in \mathbb{R}$ is

$$\delta_z(x) = d(z, x) = |x - z|.$$

For example, the point function at 2 is

$$\delta_2(x) = |x - 2|.$$

It has value 0 exactly at $x = 2$, and its graph is the familiar V-shape with vertex at $(2, 0)$. Thus the point 2 is encoded by the whole distance profile $x \mapsto |x - 2|$.

Remark (Dependencies).

- Definition: Metric Space

Theorem

Theorem 2.3.22 (Point Functions Identify Points)**.** *Let (X, d) be a metric space. The map*

$$X \to \delta(X), \qquad z \mapsto \delta_z$$

is a bijection.

Go to proof.

Remark (Standard quantified statement).

$$z \mapsto \delta_z \quad \text{is a bijection from } X \text{ onto } \delta(X).$$

Remark (Interpretation). No two different points have the same distance profile. The zero-distance law recovers z from δ_z, because $\delta_z(z) = 0$ and no other point has zero distance from z.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Point Function

Theorem

Theorem 2.3.23 (Point Function Inequalities)**.** *Let (X, d) be a nonempty metric space and let $z \in X$. Then, for all $a, b \in X$,*

$$\delta_z(b) - \delta_z(a) \le d(a, b) \le \delta_z(b) + \delta_z(a),$$

and

$$\delta_z(z) = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall a, b \in X\colon \quad \delta_z(b) - \delta_z(a) \le d(a, b) \le \delta_z(b) + \delta_z(a), \qquad \delta_z(z) = 0.$$

Remark (Interpretation). Point functions mimic the metric because they are built from the metric. The right-hand inequality is the triangle inequality through z, while the left-hand inequality is the reverse triangle estimate.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Point Function

Definition — Pointlike Function

Definition 2.3.24 (Pointlike Function)**.** Let (X, d) be a nonempty metric space. A function

$$u : X \to \mathbb{R}_{\ge 0}$$

is *pointlike* on X if, for all $a, b \in X$,

$$u(a) - u(b) \le d(a, b) \le u(a) + u(b).$$

Remark (Standard quantified statement).

$$u \text{ is pointlike} \quad \iff \quad u : X \to \mathbb{R}_{\ge 0} \ \wedge\ \forall a, b \in X\colon u(a) - u(b) \le d(a, b) \le u(a) + u(b).$$

Remark (Interpretation). A pointlike function satisfies the inequalities that every point function satisfies. It behaves like a distance-from-something profile, even before a point realizing that profile has been identified.

Remark (Dependencies).

- Definition: Metric Space
- Theorem: Point Function Inequalities

Theorem

Theorem 2.3.25 (Pointlike Functions with a Zero Are Point Functions)**.** *Let* (X, d) *be a metric space and let* $u : X \to \mathbb{R}_{\geq 0}$*. Then* $u \in \delta(X)$ *if and only if* u *is pointlike on* X *and*

$$0 \in u(X).$$

In that case, there is a unique $w \in X$ *such that* $u(w) = 0$*, and*

$$u = \delta_w.$$

Go to proof.

Remark (Standard quantified statement).

$$u \in \delta(X) \iff u \text{ is pointlike on } X \wedge 0 \in u(X).$$

Remark (Interpretation). The zero value anchors a pointlike function to an actual point of X. Once $u(w) = 0$, the pointlike inequalities force $u(a) = d(w, a)$ for every $a \in X$, so u is exactly the point function δ_w.

Remark (Dependencies).

- Definition: Point Function
- Definition: Pointlike Function

2.4 Examples of Metric Spaces

Remark (Exposition). The preceding metrics become metric spaces once their domains are paired with the verified distance functions. This section records the resulting spaces as objects in their own right.

Definition — Discrete Metric Space

Definition 2.4.1 (Discrete Metric Space)**.** Let X be a nonempty set. The *discrete metric space on* X is the metric space

$$(X, d_{\text{disc}}),$$

where d_{disc} is the discrete metric on X.

Remark (Standard quantified statement).

$$X \neq \varnothing \implies \text{MetricSpace}(X, d_{\text{disc}}).$$

Remark (Interpretation)*.* A discrete metric space is a space where distinct points are all separated by the same positive distance. Its geometry records equality and inequality, but no finer notion of nearness.

Remark (Dependencies)*.*

- Definition: Metric Space
- Definition: Discrete Metric
- Proposition: The Discrete Distance is a Metric

Definition — Euclidean Metric Space

Definition 2.4.2 (Euclidean Metric Space)**.** For $n \in \mathbb{N}$, the *standard Euclidean metric space* is

$$(\mathbb{R}^n, d_2),$$

where d_2 is the Euclidean metric on $\mathbb{R}^n$.

Remark (Standard quantified statement)*.*

$$n \in \mathbb{N} \Longrightarrow \mathrm{MetricSpace}(\mathbb{R}^n, d_2).$$

Remark (Interpretation)*.* The Euclidean metric space is the standard finite-dimensional distance model: points are coordinate vectors, and distance is ordinary straight-line length.

Remark (Dependencies)*.*

- Definition: Metric Space
- Definition: Euclidean Space
- Definition: Euclidean Distance on $\mathbb{R}^n$
- Corollary: Euclidean Metric

Definition — Taxicab Metric Space

Definition 2.4.3 (Taxicab Metric Space)**.** For $n \in \mathbb{N}$, the *taxicab metric space* is

$$(\mathbb{R}^n, d_1),$$

where d_1 is the taxicab metric on $\mathbb{R}^n$.

Remark (Standard quantified statement)*.*

$$n \in \mathbb{N} \Longrightarrow \mathrm{MetricSpace}(\mathbb{R}^n, d_1).$$

Remark (Interpretation)*.* The taxicab metric space has the same underlying set as Euclidean space, but distance is measured by summing coordinatewise displacements.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Taxicab Metric
- Proposition: Taxicab Metric

Definition — p-Metric Space

Definition 2.4.4 (p-Metric Space). Let $1 \leq p < \infty$ and let $n \in \mathbb{N}$. The *p-metric space* is

$$(\mathbb{R}^n, d_p),$$

where d_p is the p-metric on $\mathbb{R}^n$.

Remark (Standard quantified statement).

$$1 \leq p < \infty,\ n \in \mathbb{N} \Longrightarrow \mathrm{MetricSpace}(\mathbb{R}^n, d_p).$$

Remark (Interpretation). The p-metric spaces form a family of finite-dimensional metric spaces on the same coordinate set, with different rules for turning coordinate displacements into distance.

Remark (Dependencies).

- Definition: Metric Space
- Definition: p-Metric on $\mathbb{R}^n$
- Theorem: d_p is a Metric

Definition — Complex Metric Space

Definition 2.4.5 (Complex Metric Space). The *complex metric space* is

$$(\mathbb{C}, d_{\mathbb{C}}),$$

where $d_{\mathbb{C}}$ is the usual Euclidean distance on the complex plane.

Remark (Standard quantified statement).

$$\mathrm{MetricSpace}(\mathbb{C}, d_{\mathbb{C}}).$$

Remark (Interpretation). The complex metric space treats complex numbers as points in the Euclidean plane, while retaining the notation and algebra of $\mathbb{C}$.

Remark (Dependencies).

- Definition: Metric Space

- Definition: Complex Metric
- Proposition: The Complex Plane is a Metric Space

Definition — Bounded Sequence Metric Space

Definition 2.4.6 (Bounded Sequence Metric Space)**.** The *bounded sequence metric space* is

$$(\ell^\infty(\mathbb{R}), d_\infty),$$

where $\ell^\infty(\mathbb{R})$ is the set of bounded real sequences and d_∞ is the supremum metric.

Remark (Standard quantified statement).

$$\mathrm{MetricSpace}(\ell^\infty(\mathbb{R}), d_\infty).$$

Remark (Interpretation). The bounded sequence metric space is an infinite-dimensional example. Its distance measures the largest coordinatewise discrepancy between two bounded real sequences.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Supremum Metric on Bounded Real Sequences
- Proposition: Supremum Metric on Bounded Sequences

2.5 Distances, Boundedness, and Balls

2.5.1 Distances and Boundedness

Remark (Exposition). Before metric topology begins, the metric already supplies quantitative measurements. It measures the distance between two points, the spread of a set, the distance from a point to a set, and the separation between two sets. These distance notions are the numerical layer beneath balls, neighborhoods, closedness, limit points, and convergence.

Definition — Diameter

Definition 2.5.1 (Diameter)**.** Let (X, d) be a metric space and let $A \subseteq X$ be nonempty. The *diameter* of A is

$$\mathrm{diam}(A) := \sup\{\, d(a, b) : a, b \in A \,\},$$

whenever this supremum is finite, and is $+\infty$ otherwise.

Remark (Standard quantified statement).

$$\mathrm{diam}(A) = \sup\{\, d(a, b) : a, b \in A \,\}.$$

Remark (Interpretation). The diameter records the largest distance needed to travel between two points of A. It measures internal spread rather than location.

Remark (Dependencies).

- Definition: Metric Space

Definition — Distance from a Point to a Set

Definition 2.5.2 (Distance from a Point to a Set)**.** Let (X, d) be a metric space, let $x \in X$, and let $A \subseteq X$ be nonempty. The *distance from x to A* is

$$d(x, A) := \inf\{\, d(x, a) : a \in A \,\}.$$

Remark (Standard quantified statement).

$$d(x, A) = \inf_{a \in A} d(x, a).$$

Remark (Interpretation). The number $d(x, A)$ measures how close x can get to A. The infimum may be achieved by a point of A, but it need not be achieved in every metric space.

Remark (Dependencies).

- Definition: Metric Space

Definition — Distance Between Sets

Definition 2.5.3 (Distance Between Sets)**.** Let (X, d) be a metric space, and let $A, B \subseteq X$ be nonempty. The *distance between A and B* is

$$d(A, B) := \inf\{\, d(a, b) : a \in A,\ b \in B \,\}.$$

Remark (Standard quantified statement).

$$d(A, B) = \inf_{a \in A,\ b \in B} d(a, b).$$

Remark (Interpretation). Set distance records the smallest separation that points of the two sets can approach. Distinct sets can have distance 0 if they touch in the limit.

Remark (Dependencies).

- Definition: Metric Space

Theorem

Theorem 2.5.4 (Distance to a Union)**.** *Let (X, d) be a metric space, let $x \in X$, and let $A, B \subseteq X$ be nonempty. Then*

$$d(x, A \cup B) = \min\{d(x, A), d(x, B)\}.$$

Go to proof.

Remark (Standard quantified statement).

$$d(x, A \cup B) = \min\{d(x, A), d(x, B)\}.$$

Remark (Interpretation). To get close to a union, it is enough to get close to one of its pieces. The nearest piece determines the distance.

Remark (Dependencies).

- Definition: Distance from a Point to a Set

Theorem

Theorem 2.5.5 (Distance to an Intersection Bound). *Let (X, d) be a metric space, let $x \in X$, and let $A, B \subseteq X$ have nonempty intersection. Then*

$$d(x, A \cap B) \geq \max\{d(x, A), d(x, B)\}.$$

Go to proof.

Remark (Standard quantified statement).

$$A \cap B \neq \varnothing \implies d(x, A \cap B) \geq \max\{d(x, A), d(x, B)\}.$$

Remark (Interpretation). An intersection has fewer candidate points than either set alone. Fewer points can only make the infimum distance larger, not smaller.

Remark (Dependencies).

- Definition: Distance from a Point to a Set

Definition — Metric Bounded Set

Definition 2.5.6 (Metric Bounded Set). Let (X, d) be a metric space and let $A \subseteq X$. The set A is *bounded* if there exist $x \in X$ and $R > 0$ such that

$$d(x, a) \leq R$$

for every $a \in A$.

Remark (Standard quantified statement).

$$A \text{ is bounded} \iff \exists x \in X\, \exists R > 0\, \forall a \in A\colon d(x, a) \leq R.$$

Remark (Interpretation). A bounded set has finite spread from at least one center. The center is auxiliary: boundedness records finite reach, not a preferred location. Important trap: an infinite set can be bounded; for example, $(0, 1) \subseteq \mathbb{R}$ is bounded in the usual metric.

Remark (Examples). In $\mathbb{R}$ with the usual metric, the intervals $(0, 1)$, $[0, 1]$, (a, b), and $[a, b]$ are bounded whenever $a < b$. The rays $(0, \infty)$ and $[0, \infty)$, and the whole line $\mathbb{R}$, are not bounded.

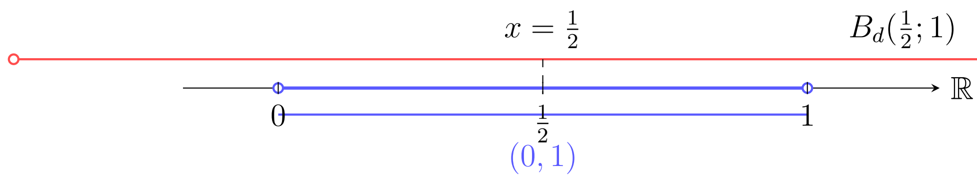
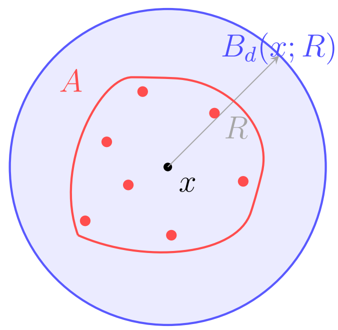


Figure 2.2: In $\mathbb{R}$, the interval $(0, 1)$ is bounded because every point of the interval is within distance 1 of $1/2$.



$A \subseteq B_d(x; R)$

Figure 2.3: An arbitrary subset A of a metric space is bounded when all of its points lie within one finite distance bound from a chosen center.

Remark (Dependencies).

- Definition: Metric Space

Theorem

Theorem 2.5.7 (Equivalent Boundedness Criterion)**.** *Let (X, d) be a metric space and let $A \subseteq X$. Then A is bounded if and only if there exist $x \in X$ and $R > 0$ such that*

$$d(x, a) < R$$

for every $a \in A$.

Remark (Standard quantified statement).

$$A \text{ is bounded} \iff \exists x \in X\, \exists R > 0\, \forall a \in A\colon d(x, a) < R.$$

Remark (Interpretation). Strict and non-strict radius bounds are equivalent after enlarging the radius. Both express the same finite-spread condition.

Remark (Dependencies).

- Definition: Metric Bounded Set

Theorem

Theorem 2.5.8 (Center Independence)**.** *Let* (X, d) *be a metric space. If* $A \subseteq X$ *and there exist* $x \in X$ *and* $R > 0$ *such that* $d(x, a) \leq R$ *for every* $a \in A$*, then for every* $y \in X$*,*

$$d(y, a) \leq R + d(x, y)$$

for every $a \in A$*. Thus boundedness does not depend on the chosen center.*

Remark (Standard quantified statement)*.*

$$\big(\forall a \in A\colon d(x, a) \leq R\big) \implies \forall y \in X \, \forall a \in A\colon d(y, a) \leq R + d(x, y).$$

Remark (Interpretation)*.* Changing centers costs at most the distance between the old center and the new center. The triangle inequality transports one finite bound to any other center.

Remark (Dependencies)*.*

- Definition: Metric Bounded Set

Theorem

Theorem 2.5.9 (Subsets of Bounded Sets Are Bounded)**.** *Let* (X, d) *be a metric space. If* $A \subseteq B \subseteq X$ *and* B *is bounded, then* A *is bounded.*

Remark (Standard quantified statement)*.*

$$A \subseteq B \subseteq X \;\wedge\; B \text{ is bounded} \implies A \text{ is bounded}.$$

Remark (Interpretation)*.* A subset cannot spread out more than the set containing it. Any bound for B automatically bounds A.

Remark (Dependencies)*.*

- Definition: Metric Bounded Set

Theorem

Theorem 2.5.10 (Finite Sets Are Bounded)**.** *Every finite subset of a metric space is bounded.*

Remark (Standard quantified statement)*.*

$$A \subseteq X \;\wedge\; A \text{ is finite} \implies A \text{ is bounded}.$$

Remark (Interpretation)*.* Only finitely many distances must be controlled. Choosing one center and a bound larger than all distances from that center captures the whole finite set.

Remark (Dependencies)*.*

- Definition: Metric Bounded Set

Theorem

Theorem 2.5.11 (Finite Unions of Bounded Sets Are Bounded)**.** *Let (X, d) be a metric space, let $n \in \mathbb{N}$, and let $A_1, \ldots, A_n \subseteq X$. If $A_1, \ldots, A_n$ are bounded, then*

$$\bigcup_{k=1}^{n} A_k$$

is bounded.

Remark (Standard quantified statement)*.*

$$\big(\forall k \in \{1, \ldots, n\} \colon A_k \text{ is bounded}\big) \Longrightarrow \bigcup_{k=1}^{n} A_k \text{ is bounded.}$$

Remark (Interpretation)*.* Each bounded set has finite spread. A finite union still has only finitely many centers and bounds to absorb into one sufficiently large bound.

Remark (Dependencies)*.*

- Definition: Metric Bounded Set
- Theorem: Center Independence

Theorem

Theorem 2.5.12 (Boundedness via Diameter)**.** *Let (X, d) be a metric space and let $A \subseteq X$ be nonempty. Then A is bounded if and only if there exists $C > 0$ such that*

$$d(a, b) \leq C$$

for all $a, b \in A$.

Remark (Standard quantified statement)*.*

$$A \neq \varnothing \Longrightarrow \Big(A \text{ is bounded} \Longleftrightarrow \exists C > 0 \, \forall a, b \in A \colon d(a, b) \leq C\Big).$$

Remark (Interpretation)*.* For nonempty sets, boundedness can be measured internally: all pairwise distances must be uniformly bounded. This is the diameter viewpoint.

Remark (Dependencies)*.*

- Definition: Metric Bounded Set
- Definition: Diameter

Theorem

Theorem 2.5.13 (Closure of a Bounded Set Is Bounded)**.** *Let (X, d) be a metric space. If $A \subseteq X$ is bounded, then $\operatorname{cl}(A)$ is bounded.*

Remark (Standard quantified statement).

$$A \text{ is bounded} \implies \mathrm{cl}(A) \text{ is bounded}.$$

Remark (Interpretation). Passing to the closure may add limit points, but it does not create points arbitrarily far away. A slightly enlarged bound still captures the closure.

Remark (Dependencies).

- Definition: Metric Bounded Set
- Definition: Metric Closure

Theorem

Theorem 2.5.14 (Compact Sets Are Bounded)**.** *Every compact subset of a metric space is bounded.*

Remark (Standard quantified statement).

$$K \subseteq X \ \wedge \ K \text{ is compact} \implies K \text{ is bounded}.$$

Remark (Interpretation). The open balls centered at any fixed point cover the compact set as their radii grow. Compactness reduces this cover to finitely many radii, and the largest one bounds the set.

Remark (Dependencies).

- Definition: Metric Bounded Set

Theorem

Theorem 2.5.15 (Totally Bounded Sets Are Bounded)**.** *Every totally bounded subset of a metric space is bounded.*

Remark (Standard quantified statement).

$$A \subseteq X \ \wedge \ A \text{ is totally bounded} \implies A \text{ is bounded}.$$

Remark (Interpretation). A totally bounded set can be covered by finitely many balls of one fixed radius. A finite collection of bounded balls has finite overall spread, so the set is bounded.

Remark (Dependencies).

- Definition: Metric Bounded Set
- Theorem: Finite Unions of Bounded Sets Are Bounded

2.5.2 Balls and Local Neighborhoods

Definition — Open Ball

Definition 2.5.16 (Open Ball)**.** Let (X, d) be a metric space, let $x \in X$, and let $r > 0$. The *open ball* of radius r centered at x is

$$B_d(x; r) := \{\, y \in X : d(y, x) < r \,\}.$$

Remark (Standard quantified statement).

$$\forall y \in X: \quad y \in B_d(x; r) \iff d(y, x) < r.$$

Remark (Interpretation). An open ball collects exactly the points whose distance from the center is strictly less than the chosen radius. The strict inequality leaves the boundary points out.

Remark (Historical note). This notation follows Definition 2.15 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Space

Definition — Closed Ball

Definition 2.5.17 (Closed Ball)**.** Let (X, d) be a metric space, let $x \in X$, and let $r > 0$. The *closed ball* of radius r centered at x is

$$\overline{B}_d(x; r) := \{\, y \in X : d(y, x) \le r \,\}.$$

Remark (Standard quantified statement).

$$\forall y \in X: \quad y \in \overline{B}_d(x; r) \iff d(y, x) \le r.$$

Remark (Interpretation). A closed ball uses the same center and radius as the corresponding open ball, but it includes the boundary points at distance exactly r.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Open Ball

Lemma

Lemma 2.5.18 (Key Local Ball Lemma)**.** *Let (X, d) be a metric space. If $y \in B_d(x; r)$, then*

$$B_d(y; r - d(x, y)) \subseteq B_d(x; r).$$

Remark (Standard quantified statement).

$$\forall x, y \in X\, \forall r > 0: \quad y \in B_d(x; r) \Longrightarrow B_d(y; r - d(x, y)) \subseteq B_d(x; r).$$

Remark (Interpretation). Every point inside an open ball has some positive-radius room that remains inside the original ball. The available radius is the unused distance from y to the boundary.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Open Ball

Theorem

Theorem 2.5.19 (Ball Nesting)**.** *Let (X, d) be a metric space, let $x \in X$, and let $0 < r \leq s$. Then*

$$B_d(x; r) \subseteq B_d(x; s).$$

If $r < s$, then also

$$\overline{B}_d(x; r) \subseteq B_d(x; s).$$

Remark (Standard quantified statement).

$$0 < r \leq s \Longrightarrow B_d(x; r) \subseteq B_d(x; s),$$

and

$$0 < r < s \Longrightarrow \overline{B}_d(x; r) \subseteq B_d(x; s).$$

Remark (Interpretation). Balls with the same center are ordered by radius. A strictly larger open ball even contains the closed ball of the smaller radius.

Remark (Dependencies).

- Definition: Open Ball
- Definition: Closed Ball

Theorem

Theorem 2.5.20 (Ball Containment by Center Distance)**.** *Let (X, d) be a metric space, let $x, y \in X$, and let $r, s > 0$.*
If

$$d(x, y) + r \leq s,$$

then

$$B_d(x; r) \subseteq B_d(y; s).$$

If

$$d(x, y) + r < s,$$

then

$$\overline{B}_d(x; r) \subseteq B_d(y; s).$$

Remark (Standard quantified statement).

$$d(x,y) + r \le s \Longrightarrow B_d(x;r) \subseteq B_d(y;s),$$

and

$$d(x,y) + r < s \Longrightarrow \overline{B}_d(x;r) \subseteq B_d(y;s).$$

Remark (Interpretation). A ball centered at x sits inside a ball centered at y when the radius available at y is large enough to pay for both the center distance and the radius around x.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Open Ball
- Definition: Closed Ball

Theorem

Theorem 2.5.21 (Intersecting Balls Imply Center-Distance Bound). *Let (X,d) be a metric space, let $x, y \in X$, and let $r, s > 0$. If*

$$B_d(x;r) \cap B_d(y;s) \ne \varnothing,$$

then

$$d(x,y) < r + s.$$

Remark (Standard quantified statement).

$$B_d(x;r) \cap B_d(y;s) \ne \varnothing \Longrightarrow d(x,y) < r + s.$$

Remark (Interpretation). If two open balls overlap, a point in the overlap gives a two-step path from one center to the other. The triangle inequality bounds the center distance by the sum of the two radii.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Open Ball

Theorem

Theorem 2.5.22 (Closed Balls Contain Open Balls). *Let (X,d) be a metric space. For every $x \in X$ and every $r > 0$,*

$$B_d(x;r) \subseteq \overline{B}_d(x;r).$$

Remark (Standard quantified statement).

$$d(x,y) < r \Longrightarrow d(x,y) \le r.$$

Remark (Interpretation). The closed ball of radius r keeps all points of the open ball and adds the possible boundary points at distance exactly r.

Remark (Dependencies).

- Definition: Open Ball
- Definition: Closed Ball

2.6 Open and Closed Sets

Remark (Exposition). Open and closed sets describe how subsets sit inside a metric space. The primitive local objects are open balls: a set is open when every one of its points has a small ball that remains inside the set. Closed sets are then defined by open complements, and closure, interior, and boundary record the points forced by these local tests.

Definition — Metric Open Set

Definition 2.6.1 (Metric Open Set). Let (X, d) be a metric space. A subset $E \subseteq X$ is *open in* (X, d) if either $E = \varnothing$ or, for every $x \in E$, there exists $\varepsilon > 0$ such that

$$B_d(x; \varepsilon) \subseteq E.$$

Remark (Standard quantified statement).

$$E \text{ is open in } (X, d) \quad \Longleftrightarrow \quad E \subseteq X \;\wedge\; \big(E = \varnothing \;\vee\; \forall x \in E\, \exists \varepsilon > 0\colon B_d(x; \varepsilon) \subseteq E\big).$$

Remark (Interpretation). An open set contains room around each of its points. The amount of room may depend on the point, but every point of the set must have some positive-radius ball that stays inside the set.

Remark (Examples). In $\mathbb{R}$ with the usual metric, each interval (a, b) is open. The interval $[a, b]$ is not open when $a < b$, because the endpoints have no positive-radius open interval contained in $[a, b]$.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Open Ball

Theorem

Theorem 2.6.2 (Open Balls are Open). *Let (X, d) be a metric space. For every $x \in X$ and every $r > 0$, the open ball $B_d(x; r)$ is open in (X, d).*
Go to proof.

Remark (Standard quantified statement).

$$\forall x \in X\, \forall r > 0\colon \quad B_d(x; r) \text{ is open in } (X, d).$$

Remark (Interpretation). Every metric ball contains room around each of its points. The triangle inequality is the mechanism that transfers room around the center to room around an arbitrary point inside the ball.

Remark (Historical note). This result corresponds to Theorem 3.4 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Space
- Definition: Open Ball
- Definition: Metric Open Set
- Lemma: Key Local Ball Lemma

Theorem

Theorem 2.6.3 (Metric Open Set Closure Properties)**.** *Let X be a metric space. Then:*

1. *the empty set $\varnothing$ and the space X are open sets;*
2. *an arbitrary union of open subsets of X is open;*
3. *any finite intersection of open subsets of X is open.*

Go to proof.

Remark (Standard quantified statement). If X is a metric space, then

$$\varnothing \text{ is open in } X, \qquad X \text{ is open in } X,$$

$$\{U_\alpha\}_{\alpha\in A} \text{ open in } X \Longrightarrow \bigcup_{\alpha\in A} U_\alpha \text{ is open in } X,$$

and, for $n \in \mathbb{N}$,

$$U_1, \ldots, U_n \text{ open in } X \Longrightarrow \bigcap_{k=1}^{n} U_k \text{ is open in } X.$$

Remark (Interpretation). Metric open sets are stable under the same basic operations that define a topology: arbitrary unions and finite intersections. Thus every metric determines a topology by declaring its metric-open subsets to be open.

Remark (Historical note). This result corresponds to Theorem 3.2 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Space
- Definition: Metric Open Set

Definition — Metric Closed Set

Definition 2.6.4 (Metric Closed Set)**.** Let (X, d) be a metric space. A subset $E \subseteq X$ is *closed in* (X, d) if its complement $E^c = X \setminus E$ is open in (X, d).

Remark (Standard quantified statement).

$$E \text{ is closed in } (X,d) \iff E \subseteq X \;\wedge\; E^c = X \setminus E \;\wedge\; E^c \text{ is open in } (X,d).$$

Remark (Interpretation). A closed set is defined by the openness of what lies outside it. In metric spaces this means every point outside E has some positive-radius ball that misses E.

Remark (Notation). When E is a subset of a fixed ambient space X, the symbol E^c denotes the complement of E in X:

$$E^c = X \setminus E.$$

Remark (Examples). In $\mathbb{R}$ with the usual metric, each interval $[a, b]$ is closed. The interval (a, b) is not closed when $a < b$, because it omits its endpoints.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Metric Open Set

Theorem

Theorem 2.6.5 (Complement of a Closed Ball Is Open)**.** *Let (X, d) be a metric space. For every $x \in X$ and every $r \geq 0$,*

$$X \setminus \overline{B}_d(x; r) = \{\, y \in X : d(x, y) > r \,\}$$

is open.

Remark (Standard quantified statement).

$$X \setminus \overline{B}_d(x; r) \text{ is open in } (X, d).$$

Remark (Interpretation). The complement of a closed ball is the region strictly beyond its radius. Every point there has a positive margin from the boundary, so a small ball around it stays in the complement.

Remark (Dependencies).

- Definition: Open Ball
- Definition: Closed Ball
- Definition: Metric Open Set

Theorem

Theorem 2.6.6 (Closed Balls Are Closed)**.** *Let (X, d) be a metric space. For every $x \in X$ and every $r \geq 0$, the closed ball*

$$\overline{B}_d(x; r) = \{\, y \in X : d(x, y) \leq r \,\}$$

is closed.

Remark (Standard quantified statement)*.*

$$\forall x \in X\, \forall r \geq 0\colon \quad \overline{B}_d(x; r) \text{ is closed in } (X, d).$$

Remark (Interpretation)*.* Closed balls are closed because their complements are open. Points outside a closed ball have a positive margin beyond the radius, and that margin gives an open ball disjoint from the closed ball.

Remark (Dependencies)*.*

- Definition: Closed Ball
- Definition: Metric Closed Set
- Theorem: Complement of a Closed Ball Is Open

Definition — Metric Closure

Definition 2.6.7 (Metric Closure)**.** Let (X, d) be a metric space and let $E \subseteq X$. The *closure* of E, denoted $\mathrm{cl}(E)$, is

$$\mathrm{cl}(E) = \{x \in X : \forall \varepsilon > 0,\ B_d(x; \varepsilon) \cap E \neq \varnothing\}.$$

Remark (Standard quantified statement)*.*

$$x \in \mathrm{cl}(E) \iff \forall \varepsilon > 0\colon B_d(x; \varepsilon) \cap E \neq \varnothing.$$

Remark (Interpretation)*.* The closure of E contains the points of E and all points that cannot be separated from E by any positive-radius ball.

Remark (Dependencies)*.*

- Definition: Metric Space
- Definition: Open Ball

Theorem

Theorem 2.6.8 (Closure of an Open Ball Is Contained in the Closed Ball)**.** *Let (X, d) be a metric space. For every $x \in X$ and every $r > 0$,*

$$\mathrm{cl}\big(B_d(x; r)\big) \subseteq \overline{B}_d(x; r).$$

Remark (Standard quantified statement).

$$z \in \mathrm{cl}\big(B_d(x;r)\big) \Longrightarrow d(x,z) \le r.$$

Remark (Interpretation). Taking the closure of an open ball may add boundary points, but it cannot add points whose distance from the center is larger than the radius. Important trap: equality need not hold in every metric space,

$$\mathrm{cl}\big(B_d(x;r)\big) \ne \overline{B}_d(x;r),$$

and can occur. For example, in a discrete metric space, the closure of an open ball may be strictly smaller than the corresponding closed ball.

Remark (Dependencies).

- Definition: Open Ball
- Definition: Closed Ball
- Definition: Metric Closure

Definition — Dense Subset

Definition 2.6.9 (Dense Subset)**.** Let (X,d) be a metric space and let $A \subseteq X$. The set A is *dense in* X if

$$\mathrm{cl}(A) = X.$$

Remark (Standard quantified statement).

$$A \text{ is dense in } X \quad \Longleftrightarrow \quad \mathrm{cl}(A) = X.$$

Remark (Interpretation). A dense set reaches every point of the ambient space by approximation. Every point of X either belongs to A or is arbitrarily close to points of A.

Remark (Examples). In $\mathbb{R}$ with the usual metric, $\mathbb{Q}$ is dense in $\mathbb{R}$, and $\mathbb{R}\setminus\mathbb{Q}$ is also dense in $\mathbb{R}$.

Remark (Dependencies).

- Definition: Metric Closure

Definition — Metric Interior Point

Definition 2.6.10 (Metric Interior Point)**.** Let (X,d) be a metric space, let $E \subseteq X$, and let $x \in X$. The point x is an *interior point of* E if there exists $\delta > 0$ such that

$$B_d(x;\delta) \subseteq E.$$

Remark (Standard quantified statement).

$$x \text{ is an interior point of } E \quad \Longleftrightarrow \quad x \in X \ \wedge\ \exists \delta > 0\colon B_d(x;\delta) \subseteq E.$$

Remark (Interpretation)*.* An interior point of E is a point with positive-radius room inside E. The entire ball around the point must remain in E, not merely the point itself.

Remark (Dependencies)*.*

- Definition: Metric Space
- Definition: Open Ball

Definition — Metric Interior

Definition 2.6.11 (Metric Interior)**.** Let (X, d) be a metric space and let $E \subseteq X$. The *interior of* E is the set of all interior points of E. It is denoted by

$$E^\circ.$$

Remark (Standard quantified statement)*.*

$$E^\circ = \{x \in X : \exists\delta > 0 \text{ such that } B_d(x; \delta) \subseteq E\}.$$

Remark (Interpretation)*.* The interior E° collects exactly the points of E that have positive-radius room inside E.

Remark (Notation)*.* The notation E° is read as the *interior of* E or E*-interior*.

Remark (Dependencies)*.*

- Definition: Metric Space
- Definition: Metric Interior Point

Theorem

Theorem 2.6.12 (Interior is the Largest Open Subset)**.** *Let E be a subset of a metric space X. Then E° is the largest open subset of X contained in E. In other words:*

1. $E^\circ \subseteq E$;
2. E° *is open;*
3. $E^\circ \supseteq O$, *for every open set* $O \subseteq E$.

Go to proof.

Remark (Standard quantified statement)*.*

$$E^\circ \subseteq E, \qquad E^\circ \text{ is open in } X, \qquad \forall O \subseteq E\colon O \text{ open in } X \implies O \subseteq E^\circ.$$

Remark (Interpretation)*.* The interior operation extracts the open part of a set. It keeps exactly the points with room inside E, and it contains every open subset already lying inside E.

Remark (Historical note). This result corresponds to Theorem 3.5 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Space
- Definition: Metric Interior
- Definition: Metric Open Set
- Theorem: Open Balls are Open

Definition — Metric Boundary

Definition 2.6.13 (Metric Boundary)**.** Let (X, d) be a metric space and let $E \subseteq X$. The *boundary* of E is

$$\partial E = \operatorname{cl}(E) \cap \operatorname{cl}(X \setminus E).$$

Remark (Standard quantified statement).

$$x \in \partial E \iff x \in \operatorname{cl}(E) \;\wedge\; x \in \operatorname{cl}(X \setminus E).$$

Remark (Interpretation). Boundary points are touched from both sides: every small ball around a boundary point meets E and also meets the complement of E.

Remark (Dependencies).

- Definition: Metric Closure

2.7 Limit Points and Isolated Points

Remark (Exposition). Limit points and isolated points separate two ways a subset can contain or approach a point. A limit point is repeatedly approached by other points of the set at every scale. An isolated point belongs to the set but has a small ball in which it is the only point of the set.

Definition — Metric Limit Point and Isolated Point

Definition 2.7.1 (Metric Limit Point and Isolated Point)**.** Let E be a subset of a metric space (X, d), and let $x \in X$.

1. The point x is a *limit point of* E if, for every $\varepsilon > 0$, the ball $B_d(x; \varepsilon)$ contains a point of $E \setminus \{x\}$.
2. The point x is an *isolated point of* E if $x \in E$ and x is not a limit point of E.

Remark (Standard quantified statement).

$$x \text{ is a limit point of } E \iff \forall \varepsilon > 0\colon B_d(x; \varepsilon) \cap (E \setminus \{x\}) \neq \varnothing.$$

$$x \text{ is an isolated point of } E \iff x \in E \;\wedge\; \exists \varepsilon > 0\colon B_d(x; \varepsilon) \cap E = \{x\}.$$

Remark (Interpretation). A limit point of E may lie in E or outside E; what matters is that points of E distinct from x occur arbitrarily close to x. An isolated point must belong to E, and some positive-radius ball around it contains no other point of E.

Remark (Notation). The set of limit points of E is denoted by E', read as E-*prime*.

Remark (Exposition). Limit points are also called *cluster points* or *accumulation points*. In later work with functions on subsets of metric spaces, limits of the form $\lim_{y\to x} f(y)$ are only meaningful when x is a limit point of the domain being approached.

Remark (Examples). In the usual metric on $\mathbb{R}$, if

$$E = (0,1) \cup \{2\},$$

then 2 is an isolated point of E, while $[0,1]$ is the set of limit points of E.

There are also metric spaces with infinite subsets that have no limit points. For example, if X is an infinite discrete metric space and $E = X$, then $E' = \varnothing$. Also, in $X = \mathbb{R} \setminus \mathbb{Q}$, the set

$$E = \{\sqrt{2}/n : n \in \mathbb{N}\}$$

has no limit point in X.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Open Ball

Theorem

Theorem 2.7.2 (Closed Sets Contain Their Limit Points). *Let X be a metric space and let $E \subseteq X$. Then E is closed if and only if $E' \subseteq E$.*

Go to proof.

Remark (Standard quantified statement).

$$E \text{ is closed in } X \iff E' \subseteq E.$$

Remark (Interpretation). Closed sets contain all points that can be approached by other points of the set. Equivalently, no limit point of a closed set is missing from the set.

Remark (Historical note). This result corresponds to Theorem 3.10 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Closed Set
- Definition: Metric Limit Point and Isolated Point

Theorem

Theorem 2.7.3 (Sequential Characterization of Limit Points)**.** *Let (X,d) be a metric space, let $E \subseteq X$, and let $x \in X$. Then $x \in E'$ if and only if there exists a sequence of distinct terms in E that converges to x.*

Go to proof.

Remark (Standard quantified statement).

$$x \in E' \iff \exists \{x_n\} \subseteq E\colon \big(\forall n \neq m,\ x_n \neq x_m\big) \wedge x_n \to x.$$

Remark (Interpretation). Limit points can be detected sequentially. A point is a limit point exactly when the set supplies infinitely many distinct approximating terms converging to that point.

Remark (Historical note). This result corresponds to Theorem 3.11 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Limit Point and Isolated Point
- Definition: Metric Sequential Convergence

Corollary

Corollary 2.7.4 (Neighborhood Characterization of Limit Points)**.** *Let (X,d) be a metric space, let $E \subseteq X$, and let $x \in X$. Then $x \in E'$ if and only if every neighborhood of x contains infinitely many points of E.*

Go to proof.

Remark (Standard quantified statement).

$$x \in E' \iff \forall U\colon \big(U \text{ is a neighborhood of } x\big) \implies U \cap E \text{ is infinite.}$$

Remark (Interpretation). The neighborhood form strengthens the ball definition into a count statement: near a limit point, every neighborhood contains not just one nearby point of E, but infinitely many.

Remark (Historical note). This result corresponds to Corollary 3.12 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Neighborhood
- Theorem: Sequential Characterization of Limit Points

Theorem

Theorem 2.7.5 (Bolzano–Weierstrass for Bounded Infinite Subsets)**.** *Every bounded infinite subset of $\mathbb{R}^m$ has a limit point in $\mathbb{R}^m$.*

Go to proof.

Remark (Standard quantified statement).

$$E \subseteq \mathbb{R}^m, \quad E \text{ bounded}, \quad E \text{ infinite} \quad \Longrightarrow \quad \exists x \in \mathbb{R}^m \colon x \in E'.$$

Remark (Interpretation). In finite-dimensional Euclidean space, boundedness prevents an infinite set from spreading out completely. Some point of the ambient space must be approached by infinitely many points of the set.

Remark (Historical note). This result corresponds to Theorem 3.13 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Theorem: Bolzano–Weierstrass in $\mathbb{R}^m$
- Theorem: Sequential Characterization of Limit Points

Proposition

Proposition 2.7.6 (The Derived Set is Closed)**.** *The set of limit points of any set is closed.*

Go to proof.

Remark (Standard quantified statement).

$$E \subseteq X \quad \Longrightarrow \quad E' \text{ is closed in } X.$$

Remark (Interpretation). Taking limit points produces a closed set. Once a point is approached by limit points of E, it is itself forced to be a limit point of E.

Remark (Historical note). This result corresponds to Proposition 3.15 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Limit Point and Isolated Point
- Theorem: Closed Sets Contain Their Limit Points

2.8 Sequential Convergence

Definition — Metric Sequential Convergence

Definition 2.8.1 (Metric Sequential Convergence)**.** Let (X, d) be a metric space. A sequence $\{x_n\}$ in X is *convergent in X* if there exists $x_0 \in X$ such that for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$d(x_n, x_0) < \varepsilon \qquad \text{for all } n \geq N.$$

In this case, $\{x_n\}$ converges to x_0, written

$$x_n \to x_0, \qquad x_0 = \lim_{n\to\infty} x_n.$$

Remark (Standard quantified statement).

$$x_n \to x_0 \iff x_0 \in X \ \wedge\ \forall \varepsilon > 0\, \exists N \in \mathbb{N}\, \forall n \in \mathbb{N}\colon$$
$$n \geq N \implies d(x_n, x_0) < \varepsilon.$$

Remark (Interpretation). A sequence converges to x_0 when its tail eventually lies inside every metric tolerance around x_0. The metric supplies the meaning of "within ε" in the ambient space X.

Remark (Historical note). This convention follows Definition 2.20 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Space

Definition — Metric Neighborhood

Definition 2.8.2 (Metric Neighborhood)**.** Let (X, d) be a metric space and let $x \in X$. A subset $U \subseteq X$ is a *neighborhood of* x if there exists $\delta > 0$ such that

$$B_d(x; \delta) \subseteq U.$$

Remark (Standard quantified statement).

$$U \text{ is a neighborhood of } x \iff U \subseteq X \ \wedge\ \exists \delta > 0\colon B_d(x; \delta) \subseteq U.$$

Remark (Interpretation). A metric neighborhood of x is any subset of X that contains at least one open ball centered at x. The set U may be larger than that ball; the essential condition is that x has some positive-radius room inside U.

Remark (Historical note). This convention follows Definition 2.21 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Space
- Definition: Open Ball

Definition — Epsilon Neighborhood

Definition 2.8.3 (Epsilon Neighborhood)**.** Let (X, d) be a metric space, let $x \in X$, and let $\varepsilon > 0$. The *ε-neighborhood of* x is the open ball

$$B_d(x; \varepsilon) = \{y \in X : d(y, x) < \varepsilon\}.$$

Remark (Standard quantified statement).

$$y \in B_d(x; \varepsilon) \iff y \in X \ \wedge\ d(y, x) < \varepsilon.$$

Remark (Interpretation). An ε-neighborhood is the concrete neighborhood obtained by choosing a particular positive radius ε. It consists exactly of the points whose distance from x is less than that tolerance.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Open Ball
- Definition: Metric Neighborhood

Theorem

Theorem 2.8.4 (Metric Neighborhood Criterion for Sequential Convergence)**.** *Let (X, d) be a metric space, let $\{x_n\}$ be a sequence in X, and let $x \in X$. Then*

$$x_n \to x$$

if and only if every neighborhood of x contains all but finitely many terms of $\{x_n\}$.
Go to proof.

Remark (Standard quantified statement).

$$x_n \to x \quad \Longleftrightarrow \quad \forall U \subseteq X\colon \big(U \text{ is a neighborhood of } x\big) \Longrightarrow \exists N \in \mathbb{N}\, \forall n \geq N\colon x_n \in U.$$

Remark (Interpretation). Sequential convergence can be tested by neighborhoods instead of numerical ε-bounds. A convergent sequence eventually remains inside every set that contains some open ball around the limit.

Remark (Historical note). This is the neighborhood characterization stated immediately after Definition 2.21 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Sequential Convergence
- Definition: Metric Neighborhood

Definition — Metric Cauchy Sequence

Definition 2.8.5 (Metric Cauchy Sequence)**.** Let (X, d) be a metric space. A sequence $\{x_n\}$ in X is *Cauchy* if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$d(x_n, x_m) < \varepsilon \qquad \text{for all } n, m \geq N.$$

Remark (Standard quantified statement).

$$\{x_n\} \text{ is Cauchy} \quad \Longleftrightarrow \quad \forall \varepsilon > 0\, \exists N \in \mathbb{N}\, \forall n, m \in \mathbb{N}\colon n, m \geq N \Longrightarrow d(x_n, x_m) < \varepsilon.$$

Remark (Interpretation). A Cauchy sequence is eventually internally stable: sufficiently late terms are close to one another, regardless of whether a limit point has already been identified in X.

Remark (Historical note). This convention follows Definition 2.22 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Space

Theorem

Theorem 2.8.6 (Metric Convergent Sequences Have Unique Limits)**.** *In metric spaces, convergent sequences have unique limits.*

Go to proof.

Remark (Standard quantified statement).

$$\forall \{x_n\} \subseteq X \, \forall x, y \in X\colon \quad \big(x_n \to x \; \wedge \; x_n \to y\big) \Longrightarrow x = y.$$

Remark (Interpretation). In a metric space, a convergent sequence cannot settle at two different points. The positive definiteness of the metric turns zero distance between candidate limits into equality of the points.

Remark (Historical note). This result corresponds to Theorem 2.23 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Space
- Definition: Metric Sequential Convergence

Theorem

Theorem 2.8.7 (Metric Cauchy Sequence with Convergent Subsequence)**.** *Let $\{x_n\}$ be a Cauchy sequence in a metric space (X, d), let $x \in X$, and let $\{x_{n_k}\}$ be a subsequence of $\{x_n\}$ such that*

$$\lim_{k \to \infty} x_{n_k} = x.$$

Then $x_n \to x$.

Go to proof.

Remark (Standard quantified statement).

$$\forall \{x_n\} \subseteq X \, \forall x \in X\colon \quad \Big(\{x_n\} \text{ is Cauchy} \\ \wedge \exists \{x_{n_k}\} \text{ subsequence of } \{x_n\}\colon x_{n_k} \to x\Big) \Longrightarrow x_n \to x.$$

Remark (Interpretation). For a Cauchy sequence, one convergent subsequence determines the behavior of the whole sequence. The Cauchy condition forces the full tail to follow any subsequence tail that has already converged.

Remark (Historical note). This result corresponds to Theorem 2.24 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Space
- Definition: Metric Cauchy Sequence
- Definition: Metric Sequential Convergence

Theorem

Theorem 2.8.8 (Coordinatewise Sequences in $\mathbb{R}^m$)**.** *Let $\{x_n\}$ be a sequence in $\mathbb{R}^m$, and let $x_0 \in \mathbb{R}^m$. Write*

$$x_n = \left(x_n^{(1)}, \ldots, x_n^{(m)}\right) \qquad \text{for all } n \in \mathbb{N} \cup \{0\}.$$

Then:

1. *$x_n \to x_0$ if and only if $x_n^{(j)} \to x_0^{(j)}$ for every $j = 1, \ldots, m$.*
2. *$\{x_n\}$ is Cauchy if and only if $\{x_n^{(j)}\}$ is Cauchy for every $j = 1, \ldots, m$.*

Go to proof.

Remark (Standard quantified statement).

$$x_n \to x_0 \iff \forall j \in \{1, \ldots, m\} \colon x_n^{(j)} \to x_0^{(j)},$$
$$\{x_n\} \text{ is Cauchy} \iff \forall j \in \{1, \ldots, m\} \colon \{x_n^{(j)}\} \text{ is Cauchy}.$$

Remark (Interpretation). In finite-dimensional Euclidean space, convergence and Cauchy behavior can be checked one coordinate at a time. The Euclidean metric packages the finitely many coordinate estimates into one distance estimate.

Remark (Historical note). This result corresponds to Theorem 2.25 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Sequential Convergence
- Definition: Metric Cauchy Sequence
- Definition: Euclidean Distance on $\mathbb{R}^n$
- Corollary: Euclidean Metric

Theorem

Theorem 2.8.9 (Euclidean Cauchy Sequences Converge)**.** *Every Cauchy sequence in* $\mathbb{R}^m$ *is convergent in* $\mathbb{R}^m$*.*

Go to proof.

Remark (Standard quantified statement).

$$\forall \{x_n\} \subseteq \mathbb{R}^m\colon \quad \{x_n\} \text{ is Cauchy} \implies \exists x \in \mathbb{R}^m\colon x_n \to x.$$

Remark (Interpretation). $\mathbb{R}^m$ contains the limits of all of its Cauchy sequences. This is the finite-dimensional Euclidean completeness property.

Remark (Historical note). This result corresponds to Theorem 2.26 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Cauchy Sequence
- Definition: Metric Sequential Convergence
- Theorem: Coordinatewise Sequences in $\mathbb{R}^m$

Theorem

Theorem 2.8.10 (Bolzano–Weierstrass in $\mathbb{R}^m$)**.** *Every bounded sequence in* $\mathbb{R}^m$ *contains a subsequence that converges in* $\mathbb{R}^m$*.*

Go to proof.

Remark (Standard quantified statement).

$$\forall \{x_n\} \subseteq \mathbb{R}^m\colon \quad \{x_n\} \text{ is bounded} \implies \exists \{x_{n_k}\} \text{ subsequence } \exists x \in \mathbb{R}^m\colon x_{n_k} \to x.$$

Remark (Interpretation). Boundedness in finite-dimensional Euclidean space is strong enough to force some subsequence to settle to a Euclidean limit.

Remark (Historical note). This result corresponds to Theorem 2.27 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Sequential Convergence
- Definition: Euclidean Distance on $\mathbb{R}^n$
- Corollary: Euclidean Metric

Chapter 3

Topology

3.1 Topology

3.2 Definitions

Definition — Topological Space

Definition 3.2.1 (Topological Space)**.** Let X be a set. A *topology* on X is a collection $\tau \subseteq \mathcal{P}(X)$ such that:

1. $\emptyset \in \tau$ and $X \in \tau$;
2. if $\{U_i\}_{i \in I}$ is any family of sets in τ, then $\bigcup_{i \in I} U_i \in \tau$;
3. if $U_1, \ldots, U_n \in \tau$, then $\bigcap_{k=1}^{n} U_k \in \tau$.

A *topological space* is a pair (X, τ), where X is a set and τ is a topology on X. The members of τ are called the *open sets* of the space.

Remark (Interpretation)*.* A topology records which subsets are treated as open. The axioms say that openness includes the impossible and total regions, is preserved under arbitrary unions, and is preserved under finite intersections.

Remark (Examples)*.* The same set may carry different topologies. If τ_1 and τ_2 are different topologies on the same set X, then (X, τ_1) and (X, τ_2) are different topological spaces because they generally declare different subsets of X to be open.

Remark (Dependencies)*.*

- Definition: Metric Open Set

Definition — Discrete Topology

Definition 3.2.2 (Discrete Topology)**.** Let X be a set. The *discrete topology* on X is the topology

$$\tau_{\mathrm{disc}} = \mathcal{P}(X).$$

Thus every subset of X is open in the topological space $(X, \tau_{\mathrm{disc}})$.

Remark (Standard quantified statement).

$$\tau_{\mathrm{disc}} = \mathcal{P}(X), \qquad U \text{ is open in } (X, \tau_{\mathrm{disc}}) \iff U \subseteq X.$$

Remark (Interpretation). The discrete topology makes the strongest possible declaration of openness: no subset is excluded. It is the topology with the most open sets on the underlying set X. Equivalently, it is the *finest* topology on X: every topology on X is contained in τ_{disc}.

Remark (Dependencies).

- Definition: Topological Space

Definition — Sierpinski Topology

Definition 3.2.3 (Sierpinski Topology)**.** Let $X = \{0, 1\}$. The *Sierpinski topology* on X is the topology

$$\tau_{\mathrm{S}} = \{\emptyset, \{1\}, X\}.$$

The topological space (X, τ_{S}) is called the *Sierpinski space*.

Remark (Standard quantified statement).

$$U \text{ is open in } (X, \tau_{\mathrm{S}}) \iff U = \emptyset \text{ or } U = \{1\} \text{ or } U = X.$$

Remark (Interpretation). The Sierpinski topology is the smallest nontrivial topology on a two-point set. It distinguishes the two points topologically: one singleton is open and the other singleton is not.

Remark (Dependencies).

- Definition: Topological Space

Definition — Cofinite Topology

Definition 3.2.4 (Cofinite Topology)**.** Let X be a set. The *cofinite topology* on X is the topology

$$\tau_{\mathrm{cof}} = \{\, U \subseteq X : X \setminus U \text{ is finite} \,\} \cup \{\emptyset\}.$$

Thus a subset of X is open exactly when it is empty or its complement in the ambient set is finite.

Remark (Standard quantified statement).

$$U \text{ is open in } (X, \tau_{\mathrm{cof}}) \iff U = \emptyset \text{ or } X \setminus U \text{ is finite}.$$

Remark (Interpretation). The cofinite topology treats almost all of X as open. A nonempty open set may omit only finitely many points, so on an infinite set any two nonempty open sets must intersect.

Remark (Dependencies).

- Definition: Topological Space

Definition — Trivial Topology

Definition 3.2.5 (Trivial Topology)**.** Let X be a set. The *trivial topology* on X is the topology

$$\tau_{\text{triv}} = \{\emptyset, X\}.$$

Thus only $\emptyset$ and X are open in the topological space (X, τ_{triv}).

Remark (Standard quantified statement).

$$\tau_{\text{triv}} = \{\emptyset, X\}, \qquad U \text{ is open in } (X, \tau_{\text{triv}}) \iff U = \emptyset \text{ or } U = X.$$

Remark (Interpretation). The trivial topology makes the weakest possible declaration of openness: only the sets required by the topology axioms are open. It is the topology with the fewest open sets on the underlying set X. Equivalently, it is the *coarsest* topology on X: it is contained in every topology on X.

Remark (Dependencies).

- Definition: Topological Space

3.3 Topological Space Subsets

Definition — Closed Subset of a Topological Space

Definition 3.3.1 (Closed Subset of a Topological Space)**.** Let (X, τ) be a topological space, and let $F \subseteq X$. The subset F is *closed in* (X, τ) if its complement

$$X \setminus F$$

is open in (X, τ).

Remark (Standard quantified statement).

$$F \text{ is closed in } (X, \tau) \iff F \subseteq X \text{ and } X \setminus F \in \tau.$$

Remark (Interpretation). Closedness is defined by openness of the complement. A subset is closed when everything outside it forms an open subset of the ambient topological space.

Remark (Examples). The empty set $\emptyset$ and the whole space X are both open and closed in every topological space (X, τ). The set X is open by the topology axioms and closed because $X \setminus X = \emptyset$ is open. The set $\emptyset$ is open by the topology axioms and closed because $X \setminus \emptyset = X$ is open.

Remark (Dependencies).

- Definition: Topological Space

Theorem

Theorem 3.3.2 (Closed-Set Characterization of Topologies)**.** *Let X be a set, and let $\mathcal{C}$ be a collection of subsets of X. Suppose that:*

1. *$\emptyset \in \mathcal{C}$ and $X \in \mathcal{C}$;*
2. *the intersection of any family of elements of $\mathcal{C}$ belongs to $\mathcal{C}$;*
3. *the union of any two elements of $\mathcal{C}$ belongs to $\mathcal{C}$.*

Then there is a unique topology τ on X whose family of closed sets is exactly $\mathcal{C}$.
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\emptyset, X \in \mathcal{C} \land \forall \mathcal{A} \subseteq \mathcal{C}\colon \bigcap_{A \in \mathcal{A}} A \in \mathcal{C}\\ &\land \forall C_1, C_2 \in \mathcal{C}\colon C_1 \cup C_2 \in \mathcal{C}\\ &\implies \exists! \tau\colon \text{Topology}(X, \tau) \land \mathcal{C} = \{F \subseteq X : X \setminus F \in \tau\}.\end{aligned}$$

Remark (Interpretation). The topology axioms can be stated dually in terms of closed sets. Open sets are closed under arbitrary unions and finite intersections, while closed sets are closed under arbitrary intersections and finite unions.

Remark (Dependencies).

- Definition: Topological Space
- Definition: Closed Subset of a Topological Space

Definition — Finer and Coarser Topologies

Definition 3.3.3 (Finer and Coarser Topologies)**.** Let τ_1 and τ_2 be topologies on a set X. The topology τ_1 is *finer than* τ_2 if

$$\tau_2 \subseteq \tau_1.$$

Equivalently, τ_2 is *coarser than* τ_1. The two topologies are *comparable* if one is finer than the other.

Remark (Standard quantified statement).

$$\tau_1 \text{ is finer than } \tau_2 \iff \tau_2 \subseteq \tau_1.$$

$$\tau_1 \text{ and } \tau_2 \text{ are comparable} \iff \tau_2 \subseteq \tau_1 \text{ or } \tau_1 \subseteq \tau_2.$$

Remark (Interpretation). A finer topology has at least as many open sets as the coarser topology. Thus it can distinguish points and subsets at least as sharply as the coarser topology.

Remark (Examples). If $\mathcal{F}_i$ is the family of closed subsets determined by τ_i, then

$$\tau_1 \text{ is finer than } \tau_2 \iff \mathcal{F}_2 \subseteq \mathcal{F}_1.$$

The discrete topology is the finest topology on a given set because every subset is open.

Remark (Dependencies).

- Definition: Topological Space
- Definition: Discrete Topology
- Definition: Closed Subset of a Topological Space

3.4 Basis

Definition — Basis for a Topology

Definition 3.4.1 (Basis for a Topology)**.** Let (X, τ) be a topological space. A collection $\mathcal{B} \subseteq \tau$ is a *basis for the topology* τ if every open set $O \in \tau$ is the union of some subcollection of $\mathcal{B}$. The elements of $\mathcal{B}$ are called *basic elements* or *basic open sets*.

Remark (Standard quantified statement).

$$\mathcal{B} \text{ is a basis for } \tau \iff \mathcal{B} \subseteq \tau \text{ and } \forall O \in \tau\, \exists \mathcal{A} \subseteq \mathcal{B} \colon O = \bigcup_{B \in \mathcal{A}} B.$$

Remark (Interpretation). A basis is a smaller supply of open sets from which all open sets can be assembled by unions. It records local building blocks for the topology.

Remark (Dependencies).

- Definition: Topological Space

Proposition

Proposition 3.4.2 (Pointwise Characterization of a Basis)**.** *Let (X, τ) be a topological space, and let $\mathcal{B} \subseteq \tau$. Then $\mathcal{B}$ is a basis for τ if and only if for every open set $O \in \tau$ and every $x \in O$, there is $B \in \mathcal{B}$ such that*

$$x \in B \subseteq O.$$

Consequently, a subset $A \subseteq X$ is open if and only if for each $x \in A$, there is $B \in \mathcal{B}$ such that

$$x \in B \subseteq A.$$

Go to proof.

Remark (Standard quantified statement).

$$\mathcal{B} \text{ is a basis for } \tau \iff \forall O \in \tau\, \forall x \in O\, \exists B \in \mathcal{B} \colon x \in B \subseteq O.$$

$$A \in \tau \iff A \subseteq X \text{ and } \forall x \in A\, \exists B \in \mathcal{B} \colon x \in B \subseteq A.$$

Remark (Interpretation). The pointwise test says that a basis can be checked locally. Around every point of every open set, there must be a basic open set that still lies inside the original open set.

Remark (Dependencies).

- Definition: Topological Space
- Definition: Basis for a Topology

Chapter 4

Set Algebra

4.1 Set Algebras

4.2 Systems of Sets

4.2.1 Systems of Sets

Comparison of Set Systems

Property	**Semiring**	**Ring**	**Algebra**	**σ-algebra**	**Topology**
Contains $\varnothing$	✓	✓	✓	✓	✓
Contains X	—	—	✓	✓	✓
Closed under $\cap$ (finite)	✓	✓	✓	✓	✓
Closed under $\cup$ (finite)	—	✓	✓	✓	✓
Closed under $\cup$ (countable)	—	—	—	✓	—
Closed under $\cup$ (arbitrary)	—	—	—	—	✓
Closed under complement	—	—	✓	✓	—
Difference $A \setminus B$	finite disj. union	✓	✓	✓	—

The definitions and theorems of the previous sections have been about individual sets and their operations. A second level of structure emerges when one asks: which *collections* of sets are closed under given operations? This question is not abstract generality for its own sake. The three subjects that depend most heavily on this chapter — topology, measure theory, and functional analysis — each begin by specifying a particular type of collection on a given set X. A topology specifies which subsets are "open." A σ-algebra specifies which subsets are "measurable." In both cases, the definition is precisely a list of closure conditions: which set operations applied to members of the collection must produce another member.

Mastering these four structures here, as purely set-theoretic objects, means that when topology and measure theory are encountered, all the foundational work is already done. The task will be to understand the *geometric* or *analytic* meaning of the axioms, not to struggle with the set-theoretic mechanics.

Definition (Semiring of Sets)

Definition 4.2.1 (Semiring of Sets)**.** A collection $\mathcal{S}$ of subsets of a set X is a *semiring* if:

(i) $\varnothing \in \mathcal{S}$;

(ii) $A, B \in \mathcal{S}$ implies $A \cap B \in \mathcal{S}$;

(iii) for any $A, B \in \mathcal{S}$, the difference $A \setminus B$ is a *finite disjoint union* of members of $\mathcal{S}$: there exist pairwise disjoint $C_1, \dots, C_n \in \mathcal{S}$ with $A \setminus B = C_1 \sqcup \cdots \sqcup C_n$.

Remark (Dependencies).

- Subset
- Empty Set
- Intersection
- Set Difference
- Union
- Pairwise Disjoint Family
- Finite and Infinite Sets

Example (Half-open intervals form a semiring). Let $X = \mathbb{R}$ and let $\mathcal{S}$ be the collection of all half-open intervals $[a, b) = \{x \in \mathbb{R} : a \leq x < b\}$, together with $\varnothing$. Then $\mathcal{S}$ is a semiring:

- $[a, b) \cap [c, d) = [\max(a, c), \min(b, d))$, which is again a half-open interval (or $\varnothing$);
- $[a, b) \setminus [c, d)$ is either $\varnothing$, $[a, c)$, $[d, b)$, or $[a, c) \sqcup [d, b)$ — in every case, a finite disjoint union of half-open intervals.

This semiring is the natural starting point for constructing Lebesgue measure: one first defines length on $\mathcal{S}$ (as $b - a$ for $[a, b)$), then extends to the generated ring and σ-algebra.

Remark (Why semirings appear in measure theory)*.* Lebesgue measure is most naturally defined first on a semiring, because semirings have enough structure to state the countable additivity condition for a *premeasure*, yet are simple enough that explicit formulas for sets like $[a, b)$ are easy to write down. The general extension theorem (Carathéodory's theorem) then extends the premeasure to the full σ-algebra. This is why Kolmogorov & Fomin introduce semirings before algebras: they are the base case of the extension hierarchy.

Definition (Ring of Sets)

Definition 4.2.2 (Ring of Sets)**.** A collection $\mathcal{R}$ of subsets of X is a *ring of sets* if:

(i) $\varnothing \in \mathcal{R}$;

(ii) $A, B \in \mathcal{R}$ implies $A \cup B \in \mathcal{R}$;

(iii) $A, B \in \mathcal{R}$ implies $A \setminus B \in \mathcal{R}$.

Remark (Dependencies)*.*

- Subset
- Empty Set
- Union
- Set Difference

Remark (Derived closure properties of a ring)*.* A ring is closed under finitely many of each operation. Specifically, if $\mathcal{R}$ is a ring then:

- $A \cap B = A \setminus (A \setminus B) \in \mathcal{R}$ (intersection follows from set difference);
- $A \triangle B = (A \setminus B) \cup (B \setminus A) \in \mathcal{R}$ (symmetric difference follows from union and difference).

A ring is closed under all finite Boolean combinations of its elements.

Remark (Why "ring")*.* The name comes from algebra. Under the operations of symmetric difference $\triangle$ (playing the role of addition) and intersection $\cap$ (multiplication), a ring of sets is a commutative ring in the algebraic sense: $\triangle$ is commutative and associative with identity $\varnothing$, every element is its own inverse ($A \triangle A = \varnothing$), and $\cap$ distributes over $\triangle$.

Definition (Algebra of Sets)

Definition 4.2.3 (Algebra of Sets)**.** A collection $\mathcal{A}$ of subsets of X is an *algebra of sets* (or *Boolean algebra of sets*, or *field of sets*) if:

(i) $X \in \mathcal{A}$;

(ii) $A \in \mathcal{A}$ implies $A^c = X \setminus A \in \mathcal{A}$;

(iii) $A, B \in \mathcal{A}$ implies $A \cup B \in \mathcal{A}$.

Remark (Dependencies)*.*

- Subset
- Complement
- Union

Remark (Derived properties of an algebra). An algebra is a ring that also contains X. Since $\varnothing = X^c$ and $X \in \mathcal{A}$, every algebra contains $\varnothing$. Closure under finite intersection follows: $A \cap B = (A^c \cup B^c)^c \in \mathcal{A}$ by De Morgan. An algebra is therefore closed under all finite Boolean operations: finite unions, finite intersections, complements, and differences.

Definition (σ-Algebra)

Definition 4.2.4 (σ-Algebra)**.** A collection $\mathcal{M}$ of subsets of X is a *σ-algebra* (or *σ-field*) on X if:

(i) $X \in \mathcal{M}$;

(ii) $A \in \mathcal{M}$ implies $A^c \in \mathcal{M}$;

(iii) if $\{A_n\}_{n\in\mathbb{N}}$ is a countable family in $\mathcal{M}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{M}$.

The pair $(X, \mathcal{M})$ is called a *measurable space.*

Remark (Dependencies).

- Subset
- Empty Set
- Complement
- Union
- Countable Set

Remark (Derived closure properties of a σ-algebra). Every σ-algebra is an algebra (take the union of a finite family by padding with $\varnothing$). But a σ-algebra is additionally closed under countable intersections: by De Morgan,

$$\bigcap_{n=1}^{\infty} A_n = \left(\bigcup_{n=1}^{\infty} A_n^c\right)^c \in \mathcal{M}.$$

The σ in "σ-algebra" is notation for "countable" (from the German *Summe*), signalling the step up from finite to countable closure.

Remark (Why countable and not arbitrary). A σ-algebra is closed under countable unions but not arbitrary ones. This is not a deficiency — it is the right condition for measure theory. Measure is defined to be countably additive: if $\{A_n\}$ is a countable disjoint family, then $\mu\big(\bigcup A_n\big) = \sum \mu(A_n)$. Extending to uncountable unions would force every set to have either measure zero or infinite measure (a useless theory). The countability restriction is what makes the theory nontrivial.

By contrast, a topology (defined below) is closed under *arbitrary* unions but only *finite* intersections. This is a different trade-off: open sets are defined by local conditions, and any union of open sets is open because being open is a pointwise property. Countable intersections of open sets are *not* required to be open; those are a separate class called G_δ sets.

Definition (Topology)

Definition 4.2.5 (Topology)**.** A *topology* on a set X is a collection τ of subsets of X satisfying:

(i) $\varnothing \in \tau$ and $X \in \tau$;

(ii) if $\{U_\alpha\}_{\alpha \in I}$ is any family of sets in τ (indexed by any set I), then $\bigcup_{\alpha \in I} U_\alpha \in \tau$;

(iii) if $U_1, \ldots, U_n \in \tau$ (finitely many), then $U_1 \cap \cdots \cap U_n \in \tau$.

The pair (X, τ) is a *topological space*. Members of τ are called *open sets*.

Remark (Dependencies).

- Subset
- Empty Set
- Indexed Family of Sets
- Indexed Union
- Indexed Intersection
- Finite and Infinite Sets

Remark (Why arbitrary unions but only finite intersections)*.* A topology formalizes the idea of "nearness" without a distance function. The condition that arbitrary unions of open sets are open reflects the following: if a point x is near some set, and that set is a union of smaller sets, then x is near one of those smaller sets. This is a purely local condition and works for any union.

Finite intersections must be open because: if x lies in the interior of U_1 and also in the interior of U_2, it lies in the interior of $U_1 \cap U_2$. This argument works for finitely many sets (take the smallest neighborhood). For a countably infinite intersection, the "smallest neighborhood" might shrink to nothing — consider $\bigcap_{n=1}^{\infty}(-1/n, 1/n) = \{0\}$ in $\mathbb{R}$, which is not open. So infinite intersections of open sets are not required to be open.

Theorem (Intersection of σ-algebras is a σ-algebra)

Theorem 4.2.6 (Intersection of σ-algebras is a σ-algebra)**.** *Let $\{\mathcal{M}_\alpha\}_{\alpha\in I}$ be any family of σ-algebras on X (indexed by any set I). Then*

$$\mathcal{M} := \bigcap_{\alpha\in I} \mathcal{M}_\alpha$$

is a σ-algebra on X. Go to proof.

Remark (Standard quantified statement). $\big(\forall\alpha \in I,\ \mathcal{M}_\alpha$ is a σ-algebra on $X\big) \to \bigcap_{\alpha\in I}\mathcal{M}_\alpha$ is a σ-algebra Explicitly: $X \in \bigcap \mathcal{M}_\alpha$; $A \in \bigcap \mathcal{M}_\alpha \to A^c \in \bigcap \mathcal{M}_\alpha$; $\forall n,\ A_n \in \bigcap \mathcal{M}_\alpha \to \bigcup_n A_n \in \bigcap \mathcal{M}_\alpha$.

Remark (Dependencies).

- σ-Algebra
- Indexed Family of Sets
- Intersection

Remark (Proof). Check each axiom: (i) $X \in \mathcal{M}_\alpha$ for all α, so $X \in \bigcap_\alpha \mathcal{M}_\alpha$. (ii) If $A \in \mathcal{M}$ then $A \in \mathcal{M}_\alpha$ for all α, so $A^c \in \mathcal{M}_\alpha$ for all α, so $A^c \in \mathcal{M}$. (iii) If $\{A_n\} \subseteq \mathcal{M}$ then $\{A_n\} \subseteq \mathcal{M}_\alpha$ for all α, so $\bigcup A_n \in \mathcal{M}_\alpha$ for all α, so $\bigcup A_n \in \mathcal{M}$. All three conditions are verified. The same argument works for algebras, rings, and topologies (with appropriate closure conditions).

Definition (Generated σ-Algebra)

Definition 4.2.7 (Generated σ-Algebra)**.** Let $\mathcal{F}$ be any collection of subsets of X. The *σ-algebra generated by $\mathcal{F}$*, written $\sigma(\mathcal{F})$, is the smallest σ-algebra on X that contains $\mathcal{F}$:

$$\sigma(\mathcal{F}) := \bigcap \big\{ \mathcal{M} \ \big|\ \mathcal{M} \text{ is a } \sigma\text{-algebra on } X \text{ and } \mathcal{F} \subseteq \mathcal{M} \big\}.$$

The collection $\mathcal{F}$ is called a *generating family* for $\sigma(\mathcal{F})$.

Remark (Dependencies).

- σ-Algebra
- Subset
- Intersection
- Intersection of σ-Algebras

Remark (Why the definition is valid). The set of σ-algebras on X containing $\mathcal{F}$ is nonempty: $\mathcal{P}(X)$ is a σ-algebra containing every subset of X. Theorem 4.2.6 guarantees their intersection is again a σ-algebra. So $\sigma(\mathcal{F})$ is always well-defined. The same construction works for generated algebras, generated rings, and (with a small modification for topologies) generated topologies.

Remark (How to think about generated structures). $\sigma(\mathcal{F})$ contains $\mathcal{F}$, and then contains everything that must be present in any σ-algebra that contains $\mathcal{F}$: complements of elements of $\mathcal{F}$, countable unions of elements of $\mathcal{F}$, countable unions of complements of elements of $\mathcal{F}$, and so on — iterating indefinitely. The generated σ-algebra is the "closure" of $\mathcal{F}$ under all σ-algebraic operations. In practice, one specifies $\mathcal{F}$ as a convenient collection (e.g., intervals) and then works with the full σ-algebra by knowing which operations are allowed.

Definition (Borel σ-Algebra)

Definition 4.2.8 (Borel σ-Algebra)**.** Let (X, τ) be a topological space. The *Borel σ-algebra* on X, written $\mathcal{B}(X)$, is the σ-algebra generated by the open sets:

$$\mathcal{B}(X) \; := \; \sigma(\tau).$$

Members of $\mathcal{B}(X)$ are called *Borel sets.*

Remark (Dependencies).

- Topology
- Generated σ-Algebra

Example (F_σ and G_δ sets). In a topological space X:

- A G_δ *set* is a countable intersection of open sets: $A = \bigcap_{n=1}^{\infty} U_n$ with each U_n open.
- An F_σ *set* is a countable union of closed sets: $A = \bigcup_{n=1}^{\infty} F_n$ with each F_n closed.

These names come from German: G for *Gebiet* (open set), δ for *Durchschnitt* (intersection); F for *Ferme* (closed), σ for *Summe* (union). Both G_δ and F_σ sets are Borel sets. They are dual: A is G_δ if and only if A^c is F_σ.

These classes appear in analysis and measure theory as the natural "next level" above open and closed sets. In $\mathbb{R}$: the set of continuity points of a monotone function is a G_δ set; the rational numbers $\mathbb{Q}$ are an F_σ set; the irrational numbers $\mathbb{R} \setminus \mathbb{Q}$ are a G_δ set.

Remark (Transition). The four structures — semiring, ring, algebra, σ-algebra — and the topology form a hierarchy of increasing closure strength. Measure theory builds from the bottom (semiring, ring) upward; topology is its own branch with a different closure pattern. The Borel σ-algebra bridges the two: it is generated by the topology, so it captures all sets describable by topological operations, and it is the standard "default" σ-algebra in any analytic context. Whenever a later chapter says "let f be a measurable function" without specifying the σ-algebra, it almost certainly means measurability with respect to the Borel σ-algebra.

4.3 The Power Set and Characteristic Functions

4.3.1 The Power Set and Characteristic Functions

Chapter 5

Algebras of Sets

SERIES	VOLUME IV	BOOK I	CHAPTER 5
From Cantor to Ito	Mathematical Spaces	Mathematical Spaces	Algebras of Sets

5.1 Set Operations and Hierarchy

5.2 Boolean Algebra of Sets

5.3 Rings of Sets

Bridge — Rings of Sets and Boolean Rings

Concept family: the set-theoretic operations behind the algebraic Boolean-ring model.

Remark (Dictionary).

Set operation	Ring operation
$A \triangle B$	$A + B$
$A \cap B$	$A \cdot B$
$\varnothing$	0
X	1

Under this dictionary, a ring of sets corresponds to a Boolean rng, and an algebra of sets corresponds to a Boolean ring with $1 = X$.

Remark (Structure theory lives in Volume VI). For the algebraic structure theory – idempotence implies characteristic 2, which implies $-p = p$, which implies commutativity, together with Stone representation and the finite-order consequence that no Boolean ring has order 3 – see Volume VI, Section 1.9, "Boolean Rings."

5.4 Sigma Algebras

5.5 Borel Sets

Chapter 6

Measure Spaces

SERIES
From Cantor to Ito

VOLUME IV
Mathematical Spaces

BOOK I
Mathematical Spaces

CHAPTER 6
Measure Spaces

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